

NEW YORK CITY BUILDING DEPARTMENT NOTES

ALL WORK SHALL COMPLY WITH APPLICABLE SECTIONS OF THE BUILDING CODE, CITY OF NEW YORK, EFFECTIVE NOVEMBER 7, 2022 AND ALL AMENDMENTS AND RULES AND REGULATIONS OF THE DEPARTMENT OF BUILDINGS TO DATE.

- A TEST WILL BE CONDUCTED UNDER DIRECTION OF A LICENSED PROFESSIONAL ENGINEER OR ARCHITECT OR OTHER PERSON HAVING NOT LESS THAN FIVE (5) YEARS EXPERIENCE SUPERVISING THE INSTALLATION OF THE MECHANICAL SYSTEM. THE TEST WILL SHOW COMPLIANCE WITH 2022 BUILDING CODE REQUIREMENTS AS OUTLINED IN SECTION [BC 1705.1705].
- THE LICENSED PROFESSIONAL ENGINEER OR ARCHITECT OR OTHER PERSON HAVING NOT LESS THAN FIVE (5) YEARS EXPERIENCE SUPERVISING THE INSTALLATION OF MECHANICAL SYSTEMS AND CONDUCTING SUCH TESTS WILL FILE DOCUMENTATION AND REPORT OF TEST THAT THE SYSTEM COMPLIES WITH THE CONSTRUCTION DOCUMENTS AND APPLICABLE LAWS.
- SPECIAL INSPECTIONS:
 - ALL MECHANICAL SYSTEMS, EQUIPMENT AND MATERIALS INSPECTIONS SHALL BE PERFORMED IN ACCORDANCE WITH SECTION [BC 1705.21] OF THE 2022 NYC BUILDING CODE.
 - POST-INSTALLED ANCHOR INSPECTIONS SHALL BE PERFORMED IN ACCORDANCE WITH SECTION [BC 1705.37] OF THE 2022 NYC BUILDING CODE.
 - FIRE-RESISTANT PENETRATIONS AND JOINT INSPECTION SHALL BE PERFORMED IN ACCORDANCE WITH SECTION [BC 1705.17] OF THE 2022 NYC BUILDING CODE.
- SMOKE DETECTION SYSTEMS SHALL BE INSTALLED AND SEQUENCED TO FOLLOW CONTROLS OPERATIONS IN ACCORDANCE WITH THE REQUIREMENTS OF SECTION [MC 608] OF THE 2022 NYC MECHANICAL CODE.
- FIRE DAMPERS, SMOKE DAMPERS, COMBINATION FIRE-SMOKE DAMPERS AND CEILING DAMPERS LOCATED WITHIN AIR DISTRIBUTION AND SMOKE CONTROL SYSTEMS SHALL BE INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF SECTION [MC 607] OF THE 2022 NYC MECHANICAL CODE.
- ALL FIRE DAMPERS SHALL BE ACCEPTED FOR USE BY THE NEW YORK CITY DEPARTMENT OF BUILDINGS AND SHALL BE MANUFACTURED AND INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF UL 555.
- SMOKE DAMPERS SHALL BE ACCEPTED FOR USE BY THE NEW YORK CITY DEPARTMENT OF BUILDINGS AND SHALL BE MANUFACTURED AND INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF UL 555.
- COMBINATION FIRE/SMOKE DAMPERS SHALL BE ACCEPTED FOR USE BY THE NEW YORK CITY DEPARTMENT OF BUILDINGS AND SHALL BE MANUFACTURED AND INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF BOTH UL 555 AND UL 555S.
- THESE PLANS ARE APPROVED ONLY FOR THE WORK INDICATED ON THE APPLICATION SPECIFICATION SHEET. ALL OTHER MATTERS SHOWN ARE NOT TO BE RELIED UPON OR TO BE CONSIDERED AS EITHER BEING APPROVED OR IN ACCORDANCE WITH APPLICABLE CODES.
- TO THE BEST OF MY KNOWLEDGE, BELIEF AND PROFESSIONAL JUDGMENT, THESE PLANS AND SPECIFICATIONS ARE IN COMPLIANCE WITH THE NEW YORK CITY ENERGY CONSERVATION CODE.

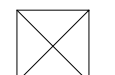
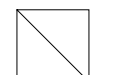
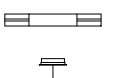
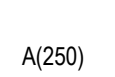
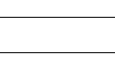
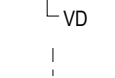
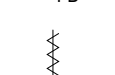
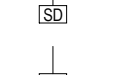
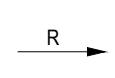
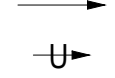
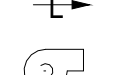
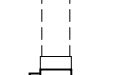
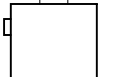
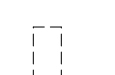
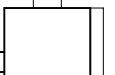
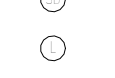
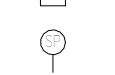
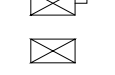
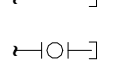
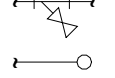
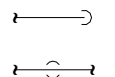
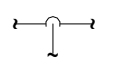
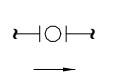
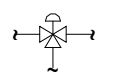
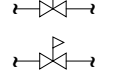
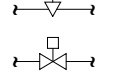
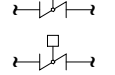
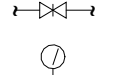
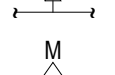
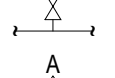
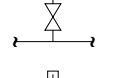
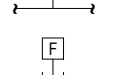
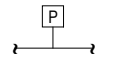
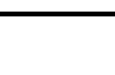
ENERGY CODE PROGRESS INSPECTIONS

A LICENSED PROFESSIONAL ENGINEER OR ARCHITECT OR OTHER PERSON HAVING NOT LESS THAN FIVE (5) YEARS EXPERIENCE SUPERVISING THE INSTALLATION OF MECHANICAL/PLUMBING SYSTEMS AND CONDUCTING SUCH TESTS WILL FILE DOCUMENTATION AND REPORT OF TEST THAT THE SYSTEM COMPLIES WITH THE CONSTRUCTION DOCUMENTS AND APPLICABLE LAWS.

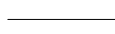
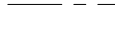
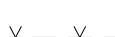
PROGRESS INSPECTIONS:

- HVAC-R AND SERVICE WATER HEATING EQUIPMENT
- HVAC-R AND SERVICE WATER HEATING SYSTEM CONTROLS
- HVAC-R AND SERVICE WATER PIPING DESIGN AND INSULATION
- DUCT LEAKAGE TESTING INSULATION AND DESIGN
- METERING
- INTERIOR LIGHTING POWER
- LIGHTING CONTROLS
- MAINTENANCE INFORMATION

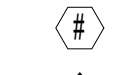
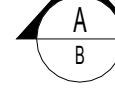
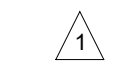
MECHANICAL LEGEND

- CEILING SUPPLY DIFFUSER
- CEILING RETURN REGISTER
- CEILING EXHAUST REGISTER
- LINEAR SUPPLY/RETURN DIFFUSER
SIDEWALL DIFFUSER
- A(250)
DIFFUSER TYPE AND CFM (CUBIC FEET PER MINUTE). REFER TO SCHEDULE.
- DOUBLE LINE DUCT
DOUBLE LINE PIPE
- ACOUSTIC LINING IN DUCT
FLEXIBLE DUCT
- WIRE MESH SCREEN
- VOLUME DAMPER
- FIRE DAMPER WITH DUCT ACCESS DOOR
- BACKDRAFT DAMPER
- MOTORIZED DAMPER WITH DUCT ACCESS DOOR
- SMOKE DAMPER WITH DUCT ACCESS DOOR
- COMBINATION FIRE/SMOKE DAMPER WITH DUCT ACCESS DOOR
- SLOPING RISE IN DUCT IN DIRECTION OF ARROW
- SLOPING DROP IN DUCT IN DIRECTION OF ARROW
- UNDERCUT DOOR
DOOR LOUVER
- FAN
- CEILING MOUNTED INLINE EXHAUST FAN
- TERMINAL BOX (VAV, CV). DESIGNATION INDICATES TYPE, BOX SIZE, AND CFM. REFER TO SCHEDULES FOR QUANTITY
- FAN POWERED TERMINAL BOX SERIES TYPE. DESIGNATION INDICATES TYPE, BOX SIZE AND CFM. REFER TO SCHEDULES FOR QUANTITY
- FAN POWERED TERMINAL BOX WITH HOT WATER REHEAT COIL. SERIES TYPE. DESIGNATION INDICATES TYPE, BOX SIZE AND CFM. REFER TO SCHEDULES FOR QUANTITY
- THERMOSTAT/TEMPERATURE SENSOR. REFER TO PLANS FOR LOCATION.
- HUMIDISTAT/HUMIDITY SENSOR. REFER TO PLANS FOR LOCATION.
- ELECTRIC ON/OFF THERMALLY PROTECTED SWITCH WITH PILOT LIGHT
- DUCT MOUNTED SMOKE DETECTOR
- SPOT TYPE LIQUID DETECTOR
- TIME CLOCK
- STATIC PRESSURE SENSOR
- STARTER/DISCONNECT SWITCH
STARTER
- CAPPED PIPE
- CAPPED AND VALVE CONNECTION (BALL VALVE UNLESS OTHERWISE NOTED)
- STRAINER WITH BLOW DOWN VALVE
- ELBOW TURNED UP
- ELBOW TURNED DOWN
- BOTTOM PIPE CONNECTION
- TOP PIPE CONNECTION
- GATE VALVE
- BALL VALVE
- CHECK VALVE
- AUTOMATIC THREE WAY VALVE
- AUTOMATIC TWO WAY CONTROL VALVE
- PRESSURE REDUCING VALVE
- PLUG VALVE
- SOLENOID VALVE
- BUTTERFLY VALVE (MANUAL)
BUTTERFLY VALVE (MOTORIZED)
- COMBINATION BALANCING VALVE AND METER STATION
- PRESSURE GAUGE WITH GATE VALVE
- MANUAL AIR VENT WITH GATE VALVE
- AUTOMATIC AIR VENT WITH GATE VALVE
- THERMOMETER
- FLOW SWITCH
- PRESSURE RETROL
- CONDENSATE PUMP (SEE SCHEDULES FOR TYPE)
- INLINE PUMP

LINE REPRESENTATION

- NEW PIPING, DUCTWORK OR EQUIPMENT
- EXISTING DUCTWORK
- EXISTING PIPING
- EXISTING PIPING, DUCTWORK OR EQUIPMENT TO BE REMOVED
- THERMOSTAT/SENSOR WIRING FROM SENSING DEVICE TO CONTROLLED DEVICE
- NEW EQUIPMENT
- EXISTING EQUIPMENT TO REMAIN
- EXISTING EQUIPMENT TO BE RELOCATED
- RELOCATED POSITION OF EXISTING EQUIPMENT
- EXISTING EQUIPMENT TO BE REMOVED

DRAWING NOTATIONS

- DRAWING NOTE TAG
- SECTION DESIGNATION ON DRAWING WHERE SECTION IS CUT
A-SECTION DESIGNATION
B-DRAWING NO.
- POINT OF NEW CONNECTION TO EXISTING WORK
- POINT OF DEMOLITION
- REMOVE AND PATCH EXISTING WORK
- REVISION SYMBOL

ABBREVIATION

- ACAIR CONDITIONING UNIT
- ACCUAIR COOLED CONDENSING UNIT
- ADACCESS DOOR
- AFAIR FILTER
- AFFABOVE FINISH FLOOR
- AHUAIR HANDLING UNIT
- ALACOUSTIC LINING
- ATCAUTOMATIC TEMPERATURE CONTROL
- AHCABOVE HUNG CEILING
- BDBACK DRAFT DAMPER
- CDCONDENSATE DRAIN
- CFMCUBIC FEET PER MINUTE
- CGCEILING GRILLE
- CWSARCONDENSER WATER SUPPLY AND RETURN
- CHWSARCHILLED WATER SUPPLY AND RETURN
- CRCEILING REGISTER
- CACCOMPUTER ROOM AIR CONDITIONING UNIT
- CWCOLD WATER
- DDRAIN
- DNDOWN
- ECMELECTRICALLY COMMUTATED MOTOR
- EDHELECTRIC DUCT HEATER
- EFEEXHAUST FAN
- FAIFRESH AIR INTAKE
- FCFLEXIBLE CONNECTION
- FCUFAN COIL UNIT
- FDFIRE DAMPER
- FP-VAVFAN POWERED VARIABLE AIR VOLUME
- FSDFIRE SMOKE DAMPER
- GCGENERAL CONTRACTOR
- HWSARHOT WATER SUPPLY AND RETURN
- KXKITCHEN EXHAUST
- LDLINEAR DIFFUSER
- LPRLOW PRESSURE RETURN
- LPSLOW PRESSURE STEAM
- MDMOTORIZED DAMPER
- MBMIXING BOX
- MODMOTORIZED OUTSIDE AIR DAMPER
- MSMOTOR STARTER
- NKNECK SIZE
- OAOBJECT AIR
- OEOBJECT ENDED DUCT
- RGRETURN GRILLE
- RARETURN AIR
- REFREFRIGERANT
- RTUROOFTOP AIR CONDITIONING UNIT
- SPSTAIR PRESSURIZATION FAN
- TOTRDUCT TRANSFER DUCT
- TFTRANSFER FAN
- TRTOP REGISTER
- TGTRANSFER GRILLE
- TXTOILET EXHAUST FAN
- UHHUNIT HEATER
- VAVVARIABLE AIR VOLUME
- VDVOLUME DAMPER (OPPOSED BLADE DAMPER)
- VFDVARIABLE FREQUENCY DRIVE
- WACWINDOW AIR CONDITIONING UNIT
- WMSWIRE MESH SCREEN

SHEET NUMBER	SHEET NAME
M-001.00	MECHANICAL LEGENDS AND NOTES
M-101.00	MECHANICAL SPECIFICATIONS SHEET 1 OF 2
M-102.00	MECHANICAL SPECIFICATIONS SHEET 2 OF 2
M-401.00	MECHANICAL SCHEDULES
M-525.00	LEVEL 29 - MECHANICAL DUCTWORK PLAN
M-530.00	LEVEL 30 - MECHANICAL DUCTWORK PLAN
M-531.00	LEVEL 31 - MECHANICAL DUCTWORK PLAN
M-529.00	LEVEL 29 - MECHANICAL PIPING PLAN
M-530.00	LEVEL 30 - MECHANICAL PIPING PLAN
M-531.00	LEVEL 31 - MECHANICAL PIPING PLAN
M-901.00	MECHANICAL DETAILS SHEET 1 OF 3
M-902.00	MECHANICAL DETAILS SHEET 2 OF 3
M-903.00	MECHANICAL DETAILS SHEET 3 OF 3
Grand total: 13	

TO THE BEST OF MY KNOWLEDGE, BELIEF AND PROFESSIONAL JUDGMENT, THESE PLANS AND SPECIFICATIONS ARE IN COMPLIANCE WITH THE 2020 NEW YORK CITY ENERGY CONSERVATION CODE.

MKDA

902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

ARCHITECT OF RECORD:

JOHN F. GIUSEFFI, ARCHITECT, PC.
902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

snowflake
7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:

AMA CONSULTING ENGINEERS
825 8TH AVE #18
NEW YORK, NY 10019
212.944.7722

AVIT/SEC CONSULTANT:

NORTHLAND CONTROLS
44150 S GRIMMER BLVD
FREMONT, CA 94538

FOOD SERVICE DESIGN CONSULTANT:

COLBURN GUYETTE
100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

DESIGN CONSULTANT:

JPC ARCHITECTS
909 112th Ave NE, Suite 206
Bellevue, WA 98004

3	07/01/2026	ISSUE FOR ADD ALTERNATE
	05/22/2026	ISSUE FOR FILING
	04/03/2026	ISSUE FOR PRICING

NO.	DATE	ISSUE
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PROJECT:

SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

MECHANICAL LEGENDS AND NOTES

SEAL & SIGNATURE	DATE:	05.22.2026
	SCALE:	N.T.S.
	PROJECT NO.	----
	DRAWN: _____	CHECKED: _____
DRAWING NO. M-001.00		

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NYC DOB NUMBER

M01397257-S1

[illegible]

BALANCE WHEEL. THE HOUSING AND WHEEL SHALL BE DESIGNED FOR QUIET LOW VELOCITY OPERATION. THE FAN HOUSING SHALL BE REMOVABLE FROM THE UNIT WITHOUT DISCONNECTING THE SUPPLY AIR DUCTWORK FOR SERVICING OF THE FAN MOTOR. THE FAN MOTOR SHALL BE AN ECM MICROPROCESSOR CONTROLLED DC TYPE MOTOR WITH INTERNAL PROGRAMMING FACTORY SET FOR THE SPECIFIC UNIT AND FEATURES SOFT START/STOP AND DELAY OFF FEATURE FOR MAXIMUM ENERGY AND QUIET OPERATION. AIR FLOW RATES SHALL BE VARIED ACCORDING TO THE STAGING OF THE UNIT. THERE WILL FURTHER BE PROVISIONS FOR ADJUSTING THE AIR DELIVERY OF THE MOTOR AIR FLOW BY +/- 15% FROM THE RATED AIRFLOW.	ELECTRICAL CONTROLS AND SAFETY DEVICES WILL BE FACTORY WIRED AND MOUNTED WITHIN THE UNIT. CONTROLS SHALL INCLUDE FAN RELAY, COMPRESSOR CONTACTOR, 24V (50 VA) TRANSFORMER, REVERSING VALVE AND SOLID STATE LOCKOUT CONTROLLER (UPM). THE UPM CONTROLLER SHALL INCLUDE THE FOLLOWING FEATURES: 1) HAND/TEST LED 2) LOW PRESSURE BYPASS TIME DELAY 3) ANTI-SHORT CYCLE TIME DELAY 4) RANDOM START TIME DELAY AND ONE TIME INTELLIGENT RESET.	XII. A LOCK-OUT INDICATING TERMINAL SHALL BE PROVIDED IN THE LOW VOLTAGE CIRCUIT. SAFETY DEVICES INCLUDE A LOW PRESSURE CUTOFF SET AT 40 PSIG FOR LOSS OF CHARGE PROTECTION AND A HIGH PRESSURE CUTOFF CONTROL SET AT 600 PSIG. THE ECM MOTOR INTERFACE BOARD SHALL PROVIDE A SCREW TYPE TERMINAL BLOCK FOR THERMOSTAT CONNECTION. LEADS TO INDICATE THERMOSTAT STATUS AND AIR DELIVERY. IT SHALL ALSO PROVIDE A MEANS OF CHANGING THE MOTOR PROGRAM TO ANY OF UP TO FOUR PREPROGRAMMED OPERATING MODES. THE THERMOSTAT HARNESS TO THE THERMOSTAT IS UNACCEPTABLE. A TERMINAL BLOCK WITH SCREW TERMINALS SHALL BE PROVIDED FOR CONTROL WIRING. A CONDENSATE OVERFLOW SWITCH SHALL BE FACTORY INSTALLED TO STOP COMPRESSOR OPERATION IF DRAIN PAN OVERFLOW IS EMINENT. AN ENERGY MANAGEMENT RELAY TO ALLOW UNIT CONTROL BY AN EXTERNAL SOURCE SHALL BE FACTORY INSTALLED.	IX. PIPING, SUPPLY, RETURN WATER AND CONDENSATE DRAIN CONNECTIONS SHALL BE BRASS FEMALE PIPE THREAD FITTINGS AND MOUNTED FLUSH TO CABINET EXTERIOR.	X. PROVIDE THE FOLLOWING ACCESSORIES: 1. HOT GAS BYPASS - HOT GAS BYPASS VALVING SHALL BE FACTORY MOUNTED AND INSTALLED INTEGRAL TO THE UNIT. THE HOT GAS BYPASS VALVE SHALL RUN OFF REFRIGERANT PRESSURE AND ALLOW UNIT TO RUN ON LOW LOAD CONDITIONS. 2. EXTRA QUIET SOUND PACKAGE - THE UNIT SHALL BE PROVIDED WITH A FACTORY BUILT AND TESTED EXTRA QUIET SOUND PACKAGE CONSISTING OF 1/2 INCH HIGH DENSITY CLOSED CELL FOAM INSULATION WRAP ON THE COMPRESSOR AND AN INSULATED DIVIDER PANEL BETWEEN THE AIR HANDLING SECTION AND THE COMPRESSOR SECTION TO MINIMIZE THE TRANSMISSION OF COMPRESSOR NOISE. ALL BULK HEADS ARE TO BE PROVIDED WITH 1" RUBATEX INSULATION. THE CABINET INTERIOR SHALL BE FINISHED WITH 1/2 INCH CONTROLLED GLASS FIBER. THE UNIT SHALL ALSO HAVE A HEAVY GAUGE STEEL COMPRESSOR BASE PAN WITH WELDED HEAVY GAGE MOUNTING RAILS FOR COMPRESSOR MOUNTING. THE COMPRESSORS ARE TO BE MOUNTED ON RESILIENT RUBBER GROMMETS FOR VIBRATION DAMPING. 3. LOW WATER TEMPERATURE PROTECTION - OPERATION DOWN TO 45F EWT IN THE COOLING MODE WITHOUT THE USE OF A WATER REGULATING VALVE CONSISTS OF: 1. AN EXTERNALLY EQUALIZED TXV FOR HEAD PRESSURE CONTROL. 2. WARMED CONDENSER 3. LOW PRESSURE TIME DELAY RELAY F. CEILING-MOUNTED DOOR CONTROL UNIT: 1. PROVIDE A SELF-CONTAINED, NON-DUCTED AIR PURIFICATION UNIT DESIGNED FOR FLUSH INSTALLATION IN A STANDARD 48" X 24" SUSPENDED T-GRID CEILING. UNIT SHALL BE CAPABLE OF DELIVERING 500-600 CFM WITH ALL CLEAN FILTERS INSTALLED AND SHALL INCLUDE AN INTEGRAL FAN ASSEMBLY, MULTI-STAGE HEATING AND COOLING, GAS-PHASE ADSORPTION MEDIA, BIPOLAR IONIZATION DEVICE, CONTROLS AND MONITORING COMPONENTS. UNIT SHALL BE SUITABLE FOR CONTROLLING PARTICULATE MATTER, ODORS, AND VOCs IN OCCUPIED SPACES SUCH AS OFFICES AND FOOD PANTRY AREAS. THE AIR PURIFICATION PROCESS SHALL INCLUDE: A. 100% HIGH EFFICIENCY HEPA FILTER, 1/2 INCH MEDIA FOR GAS-PHASE REMOVAL; A 4-INCH MERV 13 PRIMARY PARTICULATE FILTER; AND ONE (1) BIPOLAR IONIZATION MODULE. UNIT SHALL BE CONSTRUCTED OF 5052 ALUMINUM, INTERNALLY LINED WITH 1/4 INCH ACQUATEX INSULATION, AND SEALED TO PREVENT AIR BYPASS. PROVIDE WIRELESS REMOTE CONTROL, 0-10 VDC BMS FAN SPEED CONTROL, DIRTY FAN FILTER INDICATION WITH DIFFERENTIAL PRESSURE SWITCHES, AND A UNIT-MOUNTED DISCONNECT. ELECTRICAL SERVICE SHALL BE 120 OR 240 VAC, SINGLE PHASE, 60 HZ.	2.08 MOTOR STARTERS, CONTROL DEVICES AND MOTORS A. MECHANICAL CONTRACTOR TO FURNISH AND INSTALL STARTERS FOR POWER WIRING BY THE ELECTRICAL CONTRACTOR. B. MOTOR STARTERS SHALL BE CUTLER HAMMER, WESTINGHOUSE OR ALLEN-BRADLEY MANUFACTURER, SUITABLE FOR WALL OR ANGLE IRON MOUNTING, 1/2 HP TO 10 HP, 208-240V, 3 PHASE, 4 WIRE, 60 CYCLE, A.C. SERVICE. MANUAL STARTERS WITH OVERLOAD PROTECTION AND LOCKOUT TYPE DISCONNECT SWITCH OR BREAKER MAY BE USED TO CONTROL SUCH MOTORS, EXCEPT WHERE INTERLOCKS OR AUTOMATIC CONTROLS ARE REQUIRED. IN SUCH CASES, MAGNETIC ACROSS-THE-LINE STARTERS SHALL BE FURNISHED. D. ALL STARTERS FOR MOTORS 1/2 HP TO 75 HP SHALL BE COMBINATION FUSED DISCONNECT, MAGNETIC ACROSS-THE-LINE TYPE WITH FUSIBLE SWITCH. STARTERS 75 HP AND GREATER SHALL BE COMBINATION FUSED SOFT START TYPE STARTERS. E. CONTROLLERS FOR CONDENSATE PUMPS, DUPLEX AIR COMPRESSOR, PUMP AND EJECTOR PUMPS, ETC., SHALL BE FACTORY MOUNTED AND WIRED AS PART OF THE MOTOR AND AIR HANDLING SECTION. F. ALL MAGNETIC STARTERS SUBJECT TO MANUAL START AND IN DIRECT VIEW OF THE MOTORS THEY CONTROL SHALL HAVE MOMENTARY CONTACT START AND STOP BUTTONS AND PILOT LIGHT BUILT IN TO COVER. ALL MAGNETIC STARTERS SUBJECT TO ELECTRICAL INTERLOCK OR AUTOMATIC CONTROL SHALL HAVE HAND-OFF-AUTOMATIC SWITCHES AND PILOT LIGHT BUILT INTO COVER. G. WHERE STARTERS ARE NOT IN SIGHT OF MOTORS THEY CONTROL, A LOCAL DISCONNECT SWITCH WILL BE PROVIDED BY THE ELECTRICAL CONTRACTOR. H. PROVIDE ALL STARTERS WITH TRANSFORMERS BUILT INTO EACH STARTER CASING FOR CONTROL CIRCUIT. TRANSFORMERS SHALL SERVE ALL CONTROL CIRCUITS, INCLUDING INTERLOCK, INTERLOCK AND/OR AUTOMATIC CONTROL. UNIT SHALL HAVE THE NECESSARY AUXILIARY CONTACTS. ONE SET OF TERMINALS SHALL BE PROVIDED FOR EACH CONTROL CIRCUIT. CONTROL CENTERS SHALL BE PROVIDED WITH CONTROL TERMINAL BLOCKS. PROVIDE THREE SETS OF NORMALLY CLOSED OR NORMALLY OPEN CONTACTS. I. ALL MAGNETIC STARTERS SHALL HAVE THERMAL OVERLOAD IN EACH PHASE LEG AND LOW VOLTAGE PROTECTION. J. ALL PARTS SUBJECT TO WEAR, ARKING, ETC., SHALL BE REPLACEABLE. K. ALL WIRING, STARTERS, SWITCHES, ETC., SHALL BE IN FULL ACCORDANCE WITH ALL LOCAL INSURANCE UNDERWRITERS CODE REQUIREMENTS. L. FURNISH DETAILED COMPOSITE WIRING DIAGRAMS FOR THOSE INSTALLING ELECTRICAL WORK, AND FURNISH SUCH OTHER INFORMATION NECESSARY TO ASSURE THE PROPER CONNECTION, OPERATION AND CONTROL OF MOTORIZED EQUIPMENT, INCLUDING INTERLOCKS, AUTOMATIC OR SAFETY CONTROLS AND AUXILIARY CIRCUITS. M. FURNISH THE PERTINENT INFORMATION SUCH AS STARTING TIME REQUIREMENTS OF HIGH INERTIA EQUIPMENT, SO THAT THE PROPER TYPE STARTER MAY BE PROVIDED BY THE STARTER MANUFACTURER. ALL INFORMATION IS SUBJECT TO THE REVIEW OF THE ENGINEER. N. PROVIDE LAMINATED NAME PLATE ATTACHED TO EACH STARTER AND VFD IDENTIFYING THE APPLICATION IT SERVES. O. STARTERS AND VFD'S SHALL BE PROVIDED WITH ENCLOSURES RATED NEMA 1 FOR INDOOR APPLICATIONS, NEMA 3R WITH ADDITIONAL GASKETING FOR WEATHERPROOF RAINIGHT OUTDOOR ENCLOSURE OR INDOOR ENVIRONMENTS SUBJECT TO MOISTURE. P. MOTORS SHALL BE HIGH EFFICIENCY, COMPLY WITH NEMA MG-1 STANDARD AND MEET THE 1992 EPA ENERGY EFFICIENCY ACT AND UTILITY COMPANY REBATE REQUIREMENTS. Q. PROVIDE VARIABLE FREQUENCY DRIVES (VFD) AS MANUFACTURED BY GENERAL ELECTRIC, MAGNETEK, ABB, OR APPROVED EQUAL BY THE ENGINEER FOR CONTROL OF FANS AND MOTOR PUMPS AS SHOWN ON THE PLANS AND AS SPECIFIED HEREIN. VFD DISTORTION FACTOR SHALL NOT EXCEED 3% THD (RMS) AT POINT OF COMMON COUPLING, AS DEFINED BY IEEE 519-1992 AND IN NO CASE SHALL THE CURRENT THD EXCEED 10%. VFDs SHALL ACCEPT THE FOLLOWING: 1. IGBT TECHNOLOGY INCORPORATING IGBT 2. 240 CHARACTER FULL ENGLISH DIGITAL DISPLAY. CODES ARE NOT ACCEPTABLE. 3. DC LINK CHOKE 4. MANUAL BYPASS AND MANUAL BYPASS CONTACTS. 5. THREE SETS OF NORMALLY CLOSED OR NORMALLY OPEN CONTACTS. 6. CIRCUIT BREAKER DISCONNECT 7. VFD DRIVE SERVICE SWITCH 8. SPEED CONTROL DIA 9. THERMAL MOTOR OVERLOADS 10. 3% AC LINE REACTOR PRE-WIRED AND INSTALLED WITHIN VFD ENCLOSURE. 11. FACTORY START-UP SERVICE INCLUDING COMMISSION TESTING, FIELD CHECK OF CONTROL CONNECTION, AND DOCUMENTS STATING THAT ALL WORK AND DRIVE FUNCTIONS ARE DEEMED ACCEPTABLE. 12. PROGRAMMING OF ALL DRIVE PARAMETERS PARTICULAR TO THIS INSTALLATION. 13. 2 YEAR SITE PARTS AND LABOR WARRANTY AFTER START-UP. R. VARIABLE FREQUENCY DRIVE MOTORS SHALL COMPLY WITH NEMA MG-1 PART 31.4.0.2 STANDARD SUITABLE FOR VFD OPERATION. CONTRACTOR TO COORDINATE VFD AND MOTOR MANUFACTURER. S. ALL VFD DRIVES FOR ALL EQUIPMENT SHALL BE OF THE SAME MANUFACTURER. MECHANICAL CONTRACTOR SHALL COORDINATE VFD DRIVE MANUFACTURER WITH EACH EQUIPMENT VENDOR.	2.09 AUTOMATIC TEMPERATURE CONTROL A. GENERAL 1. PROVIDE ALL CONTROL, POWER, AND INTERLOCK WIRING INCLUDING CONDUITS AND INSTALL PER THE NEW YORK CITY AND NATIONAL ELECTRIC CODE. SUBMIT TERMINAL TO TERMINAL WIRING DIAGRAM, SEQUENCE OF OPERATION AND CUTS OF ALL COMPONENTS FOR APPROVAL. PROVIDE ALL RELAYS, SWITCHES, DAMPERS AND ACTUATORS, PNEUMATIC EQUIPMENT, PILOT POSITIONERS, THERMOSTATS, PANELS, LIMIT SAFETIES, TRANSFORMERS, TIME CLOCKS, CONTROL VALVES AND OTHER DEVICES TO ACCOMPLISH THE DESIRED SEQUENCE OF OPERATION. 2. FURNISH AND INSTALL AS HEREIN SPECIFIED, A COMPLETE AUTOMATIC TEMPERATURE CONTROL SYSTEM OF THE DDC TYPE WITH NATIVE BACNET DEVICES, MANUFACTURED BY SIEMENS INDUSTRY INC. 100 DEERFIELD PARKWAY, BUFFALO GROVE, IL 60089, PHONE # (312) 258-4000. THE ATC CONTRACTOR SHALL BE AN INDEPENDENT CONTRACTOR NOT AFFILIATED WITH THE MECHANICAL CONTRACTOR. 3. ALL TEMPERATURE CONTROL SYSTEMS AND COMPONENTS ARE TO BE FULLY MODULATING TYPE, EXCEPT WHERE NOTED OTHERWISE. 4. IF NEW WORK IS TO CONNECT TO AN EXISTING SYSTEM, THE PROPOSED NEW SYSTEM TO BE INSTALLED SHALL BE FULLY COMPATIBLE WITH THE EXISTING SYSTEM. THE MANUFACTURER OF THE PROPOSED NEW SYSTEM SHALL PROVIDE ALL REQUIRED INTERFACES OR GATEWAYS TO ENSURE THAT THEIR SYSTEM IS FULLY COMPATIBLE. 5. WHEN CONNECTING TO AN EXISTING BMS WORKSTATION THE CONTRACTOR SHALL UPDATE THE EXISTING BMS WORKSTATION AND GRAPHICS WITH THE NEW SYSTEMS INSTALLED/MONITORED AS PART OF THIS PROJECT. 6. TRAINING A. THE BMS CONTRACTOR SHALL PROVIDE A MINIMUM OF (8) HOURS TRAINING FOR FACILITY STAFF ON-SITE. TRAINING CAN BE PERFORMED IN SEPARATE (4) HOUR INTERVALS AT THE DISCRETION OF THE OWNER. B. THE BMS CONTRACTOR SHALL PROVIDE HARD COPY OF AS-BUILT DRAWINGS AND REVIEW ALL MAINTENANCE REQUIREMENTS AND PROCEDURES FOR ALL EQUIPMENT. 7. ALL CONTROLS MUST BE THE PRODUCT OF ONE MANUFACTURER. ALL AUTOMATIC CONTROL VALVES AND DAMPER OPERATORS SHALL BE MANUFACTURED BY THE TEMPERATURE CONTROL MANUFACTURER. 8. THE MANUFACTURER OF THE AUTOMATIC CONTROL EQUIPMENT SHALL SUBMIT THE FOLLOWING FOR APPROVAL: A SCHEMATIC DIAGRAM OF EACH CONTROL SYSTEM WHICH SHALL INDICATE THE PROPER SEQUENCE OF OPERATION AND RANGE OF THE CONTROLS FOR ALL CYCLES. PROVIDE TERMINAL POINT TO TERMINAL POINT ELECTRICAL WIRING DIAGRAMS FOR APPROVAL, A COMPLETE DESCRIPTION OF THE AUTOMATIC OPERATION OF EACH SYSTEM WHERE THE DESCRIPTION INCLUDES THE DUTY OF EACH THERMOSTAT, VALVE, SWITCH, ETC., INCORPORATED IN THE CONTROL SYSTEM WITH A SCHEDULE AND ILLUSTRATION OF ALL CONTROL INSTRUMENTS AND EQUIPMENT INCLUDING CONTROL PANELS AND DEVICES FOR EACH SYSTEM. 9. INDIVIDUAL SMOKE DETECTORS SHALL BE INSTALLED (PROVIDED BY ELECTRICAL CONTRACTOR) IN THE RETURN DUCT OF ALL AIR HANDLING SYSTEMS SHARING A COMMON CEILING OR DUCT PLENUM AS REQUIRED BY CODE. 10. FOR AIR DISTRIBUTION SYSTEMS 2,000 CFM OR LARGER, INSTALL SMOKE DETECTORS (PROVIDED BY ELECTRICAL CONTRACTOR) IN MAIN SUPPLY DUCT (DOWNSTREAM OF AIR FILTERS AND AHEAD OF ANY BRANCH CONNECTIONS) AND MAIN RETURN DUCT (UPSTREAM OF ANY FILTERS AND BEFORE RETURN AIR IS DILUTED WITH OUTDOOR AIR). 11. PROVIDE SMOKE DAMPERS (N.C.) IN THE SUPPLY (AHEAD OF ANY BRANCH CONNECTIONS) AND RETURN DUCTS FOR SUPPLY AIR SYSTEMS 1,500 CFM OR LARGER AND SERVING MORE THAN THE FLOOR ON WHICH THE SYSTEM IS LOCATED. PROVIDE END SWITCHES IN SMOKE DAMPERS FOR CONNECTION TO FIRE ALARM SYSTEM AND HVAC CONTROLS. 12. ALL SMOKE DETECTORS SHALL BE TIED TO THE BUILDING FIRE ALARM SYSTEM. A SIGNAL FROM THE BUILDING FIRE ALARM SYSTEM SHALL AUTOMATICALLY SHUT DOWN SYSTEM FANS AND CLOSE ALL ASSOCIATED SMOKE DAMPERS AND COMBINATION FIRE/SMOKE DAMPERS. SIGNAL, INTERLOCK WIRING, POWER WIRING AND FINAL CONNECTIONS WILL BE PROVIDED BY ELECTRICAL CONTRACTOR. 13. FIRE/SMOKE DAMPERS SHALL REMAIN OPEN AT ALL TIMES DURING NORMAL OPERATION AND SHALL BE CONTROLLED BY THE AUTOMATIC TEMPERATURE CONTROL SYSTEM FOR THE HVAC SYSTEM CONTROL. FIRE/SMOKE DAMPERS SHALL BE UNDER THE FULL CONTROL OF THE FIRE ALARM SYSTEM. CONTROL CONTRACTOR SHALL COORDINATE WITH THE ELECTRICAL CONTRACTOR FOR ALL CONTROL WIRING, DEVICES, AND INTERLOCK WIRING TO OPEN, CLOSE AND PROVE FIRE	SMOKE DAMPER POSITIONS. 14. ALL AUTOMATIC TEMPERATURE CONTROL SYSTEMS SHALL COMPLY WITH THE NEW YORK CITY ENERGY CONSERVATION CODE REQUIREMENTS. ALL HVAC SYSTEM CONTROLS SHALL COMPLY WITH THE FOLLOWING REQUIREMENTS: A. HEATING AND COOLING TO EACH ZONE SHALL BE CONTROLLED BY A THERMOSTAT. A MINIMUM OF ONE HUMIDITY CONTROL DEVICE SHALL BE INSTALLED PER HUMIDIFICATION OR DEHUMIDIFICATION SYSTEM. B. ALL ZONE THERMOSTAT SHALL OPERATE WITH A MINIMUM OF 5° DEADBAND BETWEEN HEATING AND COOLING WITH SETPOINT OVERLAP CAPABILITY. C. ALL ZONE THERMOSTATS SHALL BE OPERATED VIA THER
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DUCTWORK INSULATION				
DUCTWORK LOCATION	FLAME SPREAD RATING	SMOKE DEVELOPED INDEX RATING	INSULATION MEAN TEMPERATURE RATING	MINIMUM INSTALLED INSULATION VALUE
CONDITIONED SPACE	25	50	75°F	R-6
UNCONDITIONED SPACE	25	50	75°F	R-8
DUCTWORK INSULATION NOTES: 1. REFER TO MECHANICAL SPECIFICATION DRAWINGS FOR ADDITIONAL INFORMATION. 2. ALL MINIMUM REQUIRED DUCTWORK INSULATION VALUES COMPLY WITH THE 2020 NYCDEC.				

MINIMUM PIPE INSULATION						
FLUID TEMPERATURE RANGE	CONDUCTIVITY RANGE (K-VALUE)	INSULATION MEAN TEMPERATURE RATING	NOMINAL PIPE DIAMETER (IN INCHES)			
			<1	1 TO <1.5	1.5 TO <4	4 TO <8
			INSULATION THICKNESS REQUIRED (IN INCHES)			
141°F - 200°F	0.25 - 0.29	125°F	1.5	1.5	2.0	2.0
105°F - 140°F	0.21 - 0.28	100°F	1.0	1.0	1.5	1.5
40°F - 80°F	0.21 - 0.27	75°F	0.5	0.5	1.0	1.0
<40°F	0.20 - 0.26	50°F	0.5	1.0	1.0	1.5
PIPE INSULATION NOTES:						
1. REFER TO MECHANICAL SPECIFICATION DRAWINGS FOR ADDITIONAL INFORMATION.						
2. ALL MINIMUM REQUIRED PIPING INSULATION THICKNESS COMPLY WITH THE 2020 NYCDEC.						

AIR OUTLET SCHEDULE						
DESIGNATION	DESCRIPTION	NECK SIZE	FRAME SIZE	CFM RANGE	MAX HC	MANUFACTURER/ MODEL
A	PLAQUE TYPE SUPPLY DIFFUSER	8"Ø	24"x24"	0-200	<20	TTTUS/ OMNI
	PLAQUE TYPE SUPPLY DIFFUSER	10"Ø	24"x24"	201-350	<20	TTTUS/ OMNI
	PLAQUE TYPE SUPPLY DIFFUSER	12"Ø	24"x24"	351-450	<20	TTTUS/ OMNI
	PLAQUE TYPE SUPPLY DIFFUSER	14"Ø	24"x24"	451-600	<20	TTTUS/ OMNI
	PLAQUE TYPE SUPPLY DIFFUSER	15"Ø	24"x24"	601-750	<20	TTTUS/ OMNI
B	PLAQUE TYPE SUPPLY DIFFUSER	8"Ø	12"x12"	0-200	<20	TTTUS/ OMNI
C	LINEAR SUPPLY DIFFUSER	(1) 1 1/2" SLOT		0-50	22	TTTUS/ FL-15-HIT
D	LINEAR SUPPLY DIFFUSER	(2) 3" SLOT		0-37	22	TTTUS/ ML-38
F	SUPPLY REGISTER	12"x6"	14"x6"	0-200	19	TTTUS/ 300RL
	SUPPLY REGISTER	14"x6"	16"x6"	201-300	20	TTTUS/ 300RL
	SUPPLY REGISTER	20"x6"	22"x6"	301-450	23	TTTUS/ 300RL
	SUPPLY REGISTER	22"x22"	24"x24"	750-1000	<25	TTTUS/ 272RL
G	SUPPLY REGISTER	16"x4"	-	0-200	22	TTTUS/ S300FL
	SUPPLY REGISTER	22"x4"	-	201-300	23	TTTUS/ S300FL
	SUPPLY REGISTER	32"x4"	-	301-400	23	TTTUS/ S300FL
H	CEILING RETURN/ EXHAUST GRILLE	22"x22"	24"x24"	0-1570	<20	TTTUS/ 355RL
I	TRANSFER DUCT RETURN GRILLE	PER PLANS	PER PLANS	0-1200	<20	TTTUS/ 355RL
J	CEILING EXHAUST GRILLE	10"x10"	12"x12"	0-325	<20	TTTUS/ 355RL
K	CEILING EXHAUST GRILLE	12"x12"	24"x24"	0-500	<20	TTTUS/ 355FL
AIR OUTLET SCHEDULE NOTES (TYPICAL FOR EACH OUTLET): 1. INSTALL FOUR (4) WAY DIFFUSER UNLESS OTHERWISE NOTED. PROVIDE BLANK OFF BAFFLES FOR DIFFUSERS SHOWN TO HAVE 2-WAY AND 3-WAY PATTERNS. INCREASE NECK SIZES AS REQUIRED TO COMPENSATE FOR BLANKED-OFF. 2. DIFFUSERS SHALL BE SUITABLE FOR THE TYPE OF CEILING CONSTRUCTION BEING INSTALLED IN. 3. DIFFUSERS THAT SERVE AREAS WITHOUT HUNG CEILINGS SHALL BE SUITABLE FOR DUCTWORK MOUNTINGS. 4. ALL ADJUSTABLE AIR OUTLET PATTERN DEFLECTORS SHALL BE FIELD ADJUSTED TO OPTIMIZE AIR DISTRIBUTION PREVENTING DRAFT CONDITIONS. CONTRACTOR SHALL PLAN FOR A SECOND COMFORT FIELD ADJUSTMENT PER OWNER/ENGINEER DIRECTION. 5. PROVIDE CABLE OPERATED DAMPERS FOR LINEAR DIFFUSERS AND DIFFUSERS IN INACCESSIBLE CEILINGS. 6. NON-ACTIVE LENGTHS OF LINEAR DIFFUSERS TO BE USED AS RETURN. PROVIDE LIGHT SHIELDS. 7. FINISHES SHALL BE AS SPECIFIED BY THE ARCHITECT. 8. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.						

EXHAUST FAN SCHEDULE

DESIGNATION	SERVICE	CFM	STATIC PRESSURE (IN W.G.)	ELECTRICAL V/Ø/Hz	MOTOR (HP)	MCA	MOCP	DRIVE (BELT OR DIRECT)	VFD	CONTROL (SWITCH OR THERMOSTAT)	WEIGHT	MANUFACTURER/ MODEL	COMMENTS
TF-31-1	F&B PANTRY 31.61	100	0.25	115/601	1/15	1.6	15	DIRECT	NO	SWITCH	27	GREENHECK/ SO-70-VG	-
EF-30-1	DISH RM 30.13	500	0.4	115/601	1/10	4.75	15	DIRECT	NO	SWITCH	49	GREENHECK/ CSP-A1050-VG	-
EXHAUST FAN SCHEDULE NOTES (TYPICAL FOR EACH FAN): 1. PROVIDE FACTORY SUPPLIED VARIABLE SPEED SWITCH MOUNTED TO FAN CASING. 2. PROVIDE INTEGRAL DISCONNECT SWITCH. 3. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.													

BRANCH DUCT SIZING SCHEDULE

DUCT SIZE (INSIDE CLEAR DIMENSIONS)	CFM RANGE
10x6, 15x5	UP TO 175
14x6, 18x5	185-245
12x6, 16x6, 20x5	250-285
16x6, 24x5	290-335
14x6, 20x6, 26x5	340-375
16x6, 24x6	380-435
14x10, 18x6	440-515
16x10, 20x6	520-585
22x6	590-645
18x10, 24x6	650-715
20x10, 26x6	720-825
22x10, 28x6	830-895
28x12, 24x10, 32x6	900-1005

BRANCH DUCT SCHEDULE NOTE:
1. THIS SCHEDULE APPLIES TO BRANCH DUCTS THAT ARE NOT SIZED ON
2. ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.

FAN POWERED BOX SCHEDULE

DESIGNATION	BOX TYPE	CFM	FAN DATA				UNIT	BOX SIZE		PRIMARY AIRFLOW RANGE (CFM)			MINIMUM STATIC PRESS. PS (N.W.G.)	RADIATED SOUND LEVEL AT MAX CFM & PS-3" (NC)	DISCHARGE SOUND LEVEL AT MAX CFM & PS-3" (NC)	HOT WATER HEATING COIL							CAPACITY (MBH)	NO. OF ROWS	GPM	EWT (°F)	LWT (°F)	EAT (°F)	LAT (°F)	WATER Δ P (FT. HD)	MANUFACTURER/ MODEL	TYPE OF CONTROLS	REMARKS		
			ESP (N.W.G.)	V/0/Hz	HP	MCA		MOP	INLET (INCH)	OUTLET (INCH)	MIN	MAX				HEATING																			
FPB-B	SERIES	325/500	0.27	277/160	1/3	3.2	15	B	6" Ø	20.5x12.5"	100/150	325/500	100/150	0.04/0.06	13/20	11/23	-	-	-	-	-	-	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-		
FPB-HW-B	SERIES	500	0.28	277/160	1/3	3.2	15	B	8" Ø	20.5x12.5"	150	500	150	0.06	20	23	10.4	1	0.7	160	130	68	87	0.1	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-	
FPB-C	SERIES	900	0.32	277/160	1/3	3.2	15	C	10" Ø	20.5x12.5"	270	900	270	0.11	20	23	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-	
FPB-HW-C	SERIES	900	0.32	277/160	1/3	3.2	15	C	10" Ø	20.5x12.5"	270	900	270	0.11	20	23	12.2	1	0.7	160	125	68	80	0.1	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-
FPB-D	SERIES	1500	0.31	277/160	1/2	5.1	15	D	12" Ø	25x17.5"	450	1500	450	0.08	24	27	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-	
FPB-HW-D	SERIES	1500	0.31	277/160	1/2	5.1	15	D	12" Ø	25x17.5"	450	1500	450	0.08	24	27	23.3	1	1.2	160	119	68	82	0.34	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-
FPB-E	SERIES	1700	0.33	277/160	3/4	6.9	15	E	14" Ø	25x17.5"	510	1700	510	0.06	29	28	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-	
FPB-HW-E	SERIES	1700	0.33	277/160	3/4	6.9	15	E	14" Ø	25x17.5"	510	1700	510	0.06	29	28	24.9	1	1.2	160	119	68	81	0.36	-	-	-	-	-	-	-	-	TTTUS/DTFS-F	DDC	-
FAN COIL UNIT SCHEDULE:																																			
1. ALL PERFORMANCE BASED ON TESTS CONDUCTED IN ACCORDANCE WITH ASHRAE 130-2009 AND ASHRAE 88S-2008.															7. PROVIDE FACTORY MOUNTED BRACKETS AND TOGGLE DISCONNECT SWITCH.																				
2. ALL NC LEVELS DETERMINED USING ASHRAE 88S-2008 APPENDIX E.															8. PROVIDE FILTER RACK WITH TWO (2) FACTORY SUPPLIED FILTERS FOR RETURN INLET (ONE (1) FOR CONSTRUCTION USE AND ONE (1) FOR OPERATION USE).																				
3. THE MINIMUM SUPPLY CIRCUIT IMPACT (MCA) AND MAXIMUM OVER-CURRENT PROTECTION (MOP) RATINGS WERE CALCULATED IN ACCORDANCE WITH UL STANDARDS BASED ON MOTOR FULL LOAD CURRENT.															9. DUCTWORK TO SUBMIT INDIVIDUAL BOX SCHEDULE FOR REVIEW DURING THE SHOP DRAWING PROCESS. ACCESS SIDE SHALL BE INDICATED AND COORDINATED WITH THE DESIGN AND SHEETMETAL SHOP.																				
4. BATTERIES SHALL BE PROVIDED BY OTHERS, SHIPPED TO FACTORY AND FACTORY MOUNTED.															10. DRAINWATER TEMPERATURE IS BASED ON 40°F. RETURN AIR BASED ON 70°F.																				
5. FAN POWER BOX TO BE PROVIDED WITH ECM MOTOR AND MULTI-POINT CENTER AVERAGING INLET VELOCITY SENSOR CAPABLE OF OPERATING AT VELOCITIES AS LOW AS 250.															11. PROVIDE 277 VOLT TO 24 VOLT TRANSFORMER.																				
6. OPTIONAL REMOTE PWM ALLOWS FOR A 0-10V SIGNAL TO MODULATE FAN RPM TO MIN/MAX. IF BMS NEEDS TO STOP THE FAN OPTIONAL CONTROL SEQUENCE IS AVAILABLE WITH 0-2V SIGNAL ACTING AS A STOP SIGNAL AND 2.1-10V SIGNAL AS A FAN MIN/MAX SETTING. REVIEW SPECIFICATIONS FOR EXACT LOCATIONS AND CONTROLS NECESSARY.															12. U.O.N. EACH BOX SHALL HAVE A MINIMUM AIRFLOW SETPOINT OF 25% ON PRIMARY AIR INLET.																				
															13. UNITS SHALL BE CONSTRUCTED OF A MINIMUM 20-GAUGE GALVANIZED STEEL.																				
															14. SEE SPECS FOR MORE INFORMATION.																				
NOTE: ALL FAN POWERED BOXES LABEL FPB-HW ON SHEETS SHALL BE PROVIDED WITH HOT WATER HEATING COIL.																																			

WATER COOLED COMPUTER ROOM AIR CONDITIONING UNIT SCHEDULE

DESIGNATION	AREA SERVED	CAPACITY			SPACE CONDITION			AIR FLOW			INFRARED HUMIDIFIER		CONDENSER SECTION						ELECTRICAL W/Ø/Hz	UNIT FLA	UNIT MCA	UNIT MOP	(OPERATING WEIGHT (LBS) (EVAIP/ COND)	MANUFACTURER/ MODEL (EVAIP/ COND)	COMMENTS
		NOMAL TONS	TOTAL (MBH)	SENSIBLE (MBH)	DB (°F)	WB (°F)	% RH	CFM	ESP	VFD	LAT (DBWB °F)	LBSHR	FLA	DESIGNATION	GPM	MAX PD (FT)	EWT (°F)	LWT (°F)							
CRAC-29-1	IDF-29.35	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0421	5	8.2	WC03-29-1	9.0	10.7	85	95	460/360	16.5	20.6	25	205/150	ABOVE AIR/ WCH011A/ COW011B	WALL MOUNTED UNIT
CRAC-30-1	IDF-30.29	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0421	5	8.2	WC03-30-1	9.0	10.7	85	95	460/360	16.5	20.6	25	205/150	ABOVE AIR/ WCH011A/ COW011B	WALL MOUNTED UNIT
CRAC-31-1	IDF-31.44	3.0	37.5	37.5	75	62.5	50	1500	0.5	NO	57.0421	5	8.2	WC03-31-1	9.0	10.7	85	95	460/360	16.5	20.6	25	205/150	ABOVE AIR/ WCH011A/ COW011B	WALL MOUNTED UNIT
WATER COOLED COMPUTER ROOM AIR CONDITIONING UNIT SCHEDULE NOTES (TYPICAL FOR EACH UNIT)																									
1. PROVIDE SPRING TYPE VIBRATION ISOLATION HANGERS FOR EACH UNIT.																									
2. PROVIDE MC-SERIES ADVANCED MICROPROCESSOR W/ ALARMS & BACNET COMPATIBILITY.																									
3. PROVIDE FACTORY SUPPLIED COMBINATION STARTER AND DISCONNECT SWITCH.																									
4. PROVIDE AUXILIARY STAINLESS STEEL DRAIN PAN WITH LEAK DETECTION.																									
5. PROVIDE LOW ENTERING WATER TEMP KIT.																									
6. UNITS TO BE RATED FOR SYSTEM PRESSURE.																									
7. PROVIDE HOT GAS BYPASS.																									
8. FACTORY INSTALLED ACCESSORIES: CONDENSATE PUMP, RECEIVER, COMPRESSOR SOUND JACKET, FIRESTAT, SMOKE DETECTOR, HUMIDIFIER.																									
9. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.																									

PIPING FITTING SCHEDULE

PIPE MATERIAL	PIPE SIZE	JOINT TYPE	FITTING MATERIAL	FITTING CLASS
COPPER TUBING HARD DRAWN	4" & SMALLER	BRAZING	WROUGHT COPPER	450 PSIG AT 100°F TO 200°F, 150 PSIG AT 250°F
RED BRASS	ALL SIZES	THREADED	CAST BRONZE	125 AND 250 PSIG

PIPING FITTING NOTES:
1. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

AIR PURIFICATION SYSTEM SCHEDULE

DESIGNATION	SERVING	SUPPLY AIR FLOW CFM	FAN			STAGES OF FILTRATION				ELECTRICAL		UNIT WEIGHT (LBS)	UNIT DIMENSIONS (LxWxH)	MANUFACTURER/ MODEL	COMMENTS
			AMP	RPM	HP	1	2	3	4	V/Ø/Hz	FLA				
BAS-30-1.2	FOOD PREP/ SERVING PT	500	1.5	1725	1/8	1" PREFILTER	(2) 2" CARBON MEDIA	4" PRIMARY PARTICULATE FILTER	(2) BI-POLAR IONIZATION MODULES	115/160	5.0	90	48x24x12	BIOCLIMATIC/ MC 880	-
BIOCLIMATIC SYSTEM SCHEDULE NOTES: 1. UNIT SHALL BE A SELF-CONTAINED FLUSH MOUNTED SYSTEM FOR INSTALLATION INTO "T" GRID CEILING. 48"x24" MODULE 1" 2. ACOUSTICAL FOAM INSULATION 3. SYSTEM SHALL DELIVER SPECIFIED CFM WITH ALL FILTERS INSTALLED IN UNIT 4. ALL METAL TO METAL SURFACES SEALED WITH SILICONE SEALANT 5. SOUND LEVEL NOT TO EXCEED 53.5 DB @ 4.5 FT 6. FAN SHALL BE ON/OFF FORWARD CURVE WHEEL, TWIN UNITS 7. TWO (2) AIR DIFFERENTIAL PRESSURE SWITCHES FOR PREFILTER AND PRIMARY FILTER CHANGE 8. DIRTY FILTER LIGHT 9. EQUIPMENT SHALL BE ETL LISTED 10. UNIT MOUNTED CONTROLS FOR STANDALONE AND ON/OFF OPERATION 11. UNIT MANUFACTURER TO PROVIDE ONE FACTORY STARTUP AND ONE ANNUAL FOLLOW UP INSPECTION WITHIN 1 YEAR OF STARTUP 12. UNIT INSTALLATION AND ALL WIRING IS BY OTHERS 13. MEDIA SHIPPED SEPARATELY FOR FIELD INSTALLATION BY OTHERS 14. VIBRATION ISOLATORS AND MOUNTING HARDWARE BY OTHERS 15. MINIMUM 36" CLEARANCE REQUIRED FOR FILTER CHANGEOUT 16. UNITS NOT TO EXCEED 48"X24"X14" DIMENSIONS 17. ONE (1) SPARE MEDIA FILTER SET PER UNIT SHALL BE PROVIDED.															

PUMP SCHEDULE

DESIGNATION	SERVICE	TYPE	GPM	TDR (FT)	RPM	NPSH	PUMP EFFICIENCY	BHP	MOTOR HP	VFD	ELECTRICAL V/Ø/Hz	WEIGHT (LBS)	MANUFACTURER MODEL	COMMENTS
CWP-29-1.2	CONDENSER WATER	INLINE MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/160	73	GRUNDFOS/ CRE3-DOL2VFD	TWO PUMPS PRIMARY STANDBY
CWP-30-1.2	CONDENSER WATER	INLINE MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/160	73	GRUNDFOS/ CRE3-DOL2VFD	TWO PUMPS PRIMARY STANDBY
CWP-31-1.2	CONDENSER WATER	INLINE MULTISTAGE	10	35	2,845	3.85	84	-	0.5	YES	208/160	73	GRUNDFOS/ CRE3-DOL2VFD	TWO PUMPS PRIMARY STANDBY
PUMP SCHEDULE NOTES (TYPICAL FOR EACH PUMP): 1. PROVIDE SPRING TYPE VIBRATION AT PUMP. 2. PROVIDE COMBINATION DISCONNECT SWITCH AND STARTER AS REQUIRED. 3. PROVIDE DRIP PAN UNDER INLINE PUMPS WITH LEAK DETECTOR. 4. PUMPS TO BE RATED FOR 95 PSI. 5. MECHANICAL CONTRACTOR TO VERIFY SUPPLY AND RETURN DIFFERENTIAL PRESSURE OF EXISTING RISERS IN FIELD AT POINT OF CONNECTION AND SUBMIT PRESSURE READINGS WITH PUMP SHOP SUBMITTALS FOR REVIEW AND RECORD. 6. PROVIDE PERMANENT MAGNET ECM MOTOR (IES EFFICIENCY) WITH INTEGRAL VFD. 7. MECHANICAL CONTRACTOR TO PROVIDE NON OVERLOADING PUMP CURVE FOR REVIEW DURING THE SHOP DRAWING PROCESS. FOR VFD APPLICATIONS PROVIDE PARALLEL PUMP SPEED CURVES IN 10% INCREMENTS SHALL BE IN STRICT ACCORDANCE WITH THE MANUFACTURERS INSTRUCTIONS. 9. SEE EQUIPMENT SPECIFICATIONS FOR ADDITIONAL INFORMATION.														

DEHUMIDIFICATION UNIT SCHEDULE - ADD ALT

DESIGNATION

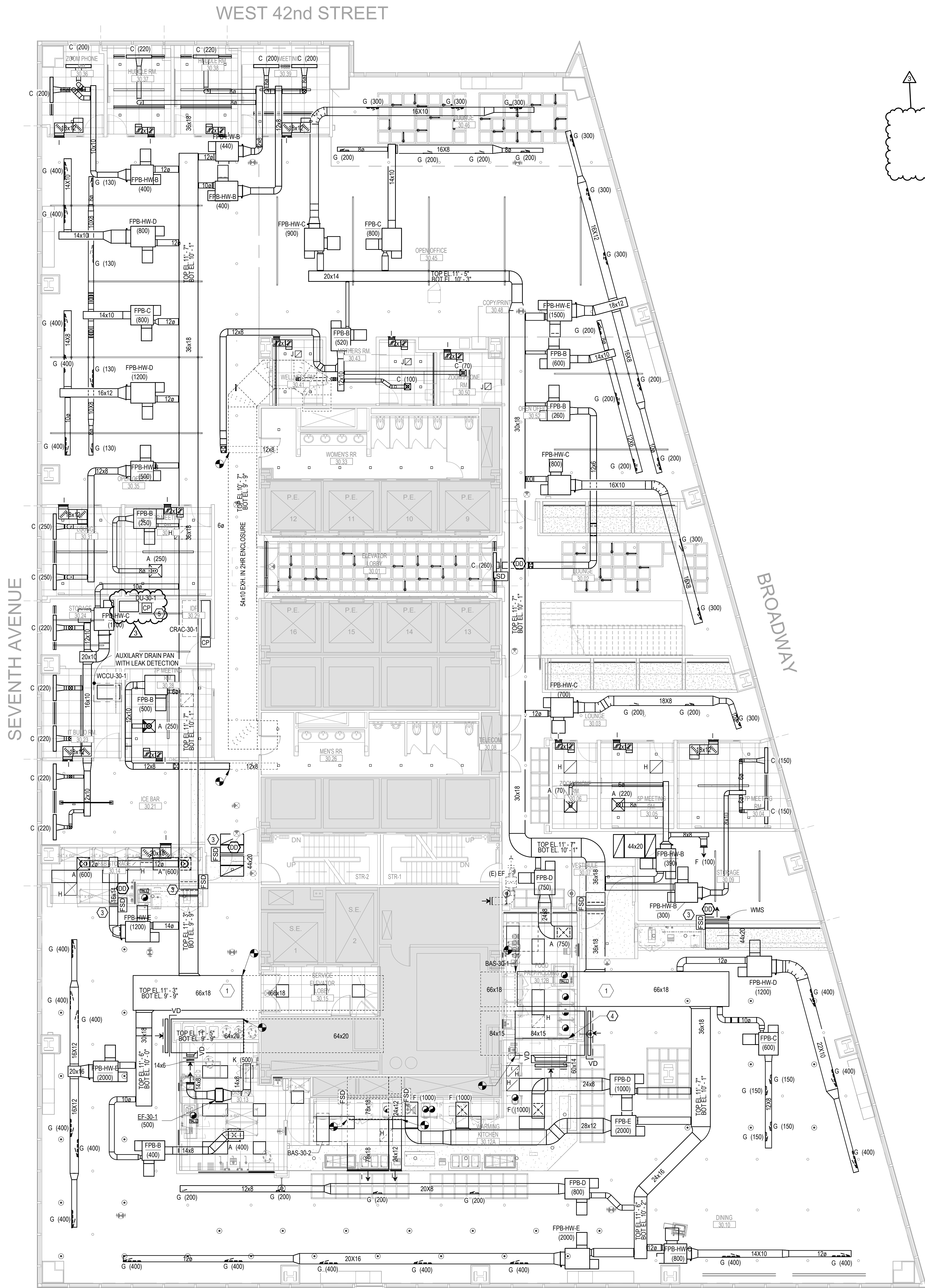
This is a detailed mechanical floor plan for the 11th floor of a building. The plan shows a complex network of HVAC ductwork, piping, and equipment. Key features include:

- Rooms and Spaces:** Labeled rooms include 'OPEN OFFICE', 'SP. MEETING RM', 'WOMEN'S RR', 'MEN'S RR', 'ELEVATOR', 'STAIRS', 'S.E.', 'S.E.', 'SERVICE', 'GREASE WASTE', 'FOCUS RM', 'ZOOM PHONE', 'DOUBLES RM', 'TRIPLE RM', 'MODULAR RM', and 'DOCKING RM'.
- Structural Elements:** Columns are labeled 'C (200)', 'C (250)', 'C (300)', 'C (400)', and 'C (500)'. Beams are labeled 'G (130)', 'G (160)', 'G (200)', 'G (300)', 'G (400)', and 'G (500)'. Walls are labeled 'W (100)', 'W (150)', 'W (200)', 'W (250)', 'W (300)', 'W (350)', 'W (400)', and 'W (450)'.
- Equipment and Piping:** Various pieces of equipment are labeled, including 'FPB-B (150)', 'FPB-B (200)', 'FPB-B (250)', 'FPB-B (300)', 'FPB-B (400)', 'FPB-B (450)', 'FPB-B (500)', 'FPB-B (600)', 'FPB-B (750)', 'FPB-B (1000)', 'FPB-B (1200)', 'FPB-B (1500)', 'FPB-B (1800)', 'FPB-B (2000)', 'FPB-B (2500)', 'FPB-B (3000)', 'FPB-B (3500)', 'FPB-B (4000)', 'FPB-B (4500)', 'FPB-B (5000)', 'FPB-B (5500)', 'FPB-B (6000)', 'FPB-B (6500)', 'FPB-B (7000)', 'FPB-B (7500)', 'FPB-B (8000)', 'FPB-B (8500)', 'FPB-B (9000)', 'FPB-B (9500)', 'FPB-B (10000)', 'FPB-B (10500)', 'FPB-B (11000)', 'FPB-B (11500)', 'FPB-B (12000)', 'FPB-B (12500)', 'FPB-B (13000)', 'FPB-B (13500)', 'FPB-B (14000)', 'FPB-B (14500)', 'FPB-B (15000)', 'FPB-B (15500)', 'FPB-B (16000)', 'FPB-B (16500)', 'FPB-B (17000)', 'FPB-B (17500)', 'FPB-B (18000)', 'FPB-B (18500)', 'FPB-B (19000)', 'FPB-B (19500)', 'FPB-B (20000)', 'FPB-B (20500)', 'FPB-B (21000)', 'FPB-B (21500)', 'FPB-B (22000)', 'FPB-B (22500)', 'FPB-B (23000)', 'FPB-B 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- 1 DUCTWORK TO BE CONSTRUCTED OF RECTANGULAR 16-GAUGE OUTER DUCT WITH ACOUSTICAL LINING, TENDON LINER AND STANDARD GAUGE PERFORATED INNER DUCT TO MATCH BUILDING STANDARD FOR FAN DISCHARGE DUCTWORK.
- 2 EXPOSED DUCTWORK SHALL BE OVAL OR ROUND SPIRAL DUCT CONSTRUCTED OF GALVANIZED STEEL WITH ACOUSTICAL LINING AS PER SPECIFICATIONS. FABRICATE AND INSTALL AS PER SMACNA STANDARDS, WITH ALL JOINTS, ELBOWS AND CONNECTIONS SUITED FOR PRESSURE CLASS, SUPPORT AND HANG DETAIL PER CODE REQUIREMENTS.
- 3 UNIT LOCATION TO BE COORDINATED WITH IDF LAYOUT. LOCATION AND POSITIONING SHOULD ALLOW FOR ALL CLEARANCES AS SPECIFIED BY MANUFACTURER. IF CEILING HUNG, UNIT SHOULD BE VIBRATION INSULATED AS DESCRIBED FOR MECHANICAL EQUIPMENT IN SPECIFICATIONS.

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KEY NOTES

- DUCTWORK TO BE CONSTRUCTED OF 16-GAUGE OUTER DUCT WITH ACOUSTICAL LINING, REGULAR LINER AND STANDARD GAUGE PERFORATED INNER DUCT TO MATCH BUILDING STANDARD FOR FAN DISCHARGE DUCTWORK.
- EXPOSED DUCTWORK SHALL BE OVAL OR ROUND SPIRAL DUCT CONSTRUCTED OF GALVANIZED STEEL WITH ACOUSTICAL LINING AS PER SPECIFICATIONS, FABRICATE AND INSTALL AS PER SMACNA STANDARDS, WITH ALL SEAMS AIRTIGHT AND CONNECTIONS SUITABLE FOR PRESSURE CLASS, SUPPORT AND HANG DUCT PER CODE REQUIREMENTS.
- PROVIDE PNEUMATIC PIPING AND WIRING AS REQUIRED FOR FIRE SMOKE DAMPER PER BUILDING RULES AND REGULATIONS. COORDINATE WITH BMS VENDOR.
- PROVIDE RETURN AIR TRANSFER DUCTS ABOVE THE CEILING WITH MINIMUM OF 1.90 SQ FEET FREE AREA AT PARTITION INDICATED. GO TO COORDINATE.
- UNIT LOCATION TO BE COORDINATED WITH IDF LAYOUT. LOCATION AND POSITIONING SHOULD ALLOW FOR ALL CLEARANCES AS SPECIFIED BY MANUFACTURER. IF CEILING HUNG, UNIT SHOULD BE VIBRATION INSULATED AS DESCRIBED FOR MECHANICAL EQUIPMENT IN SPECIFICATIONS.

MECHANICAL NOTES

- ALL WORK SHALL CONFORM TO THE LATEST BUILDING STANDARDS. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN A COPY OF THE BUILDING STANDARDS AND MEET WITH BUILDING MANAGEMENT IN ORDER TO BECOME TOTALLY FAMILIAR WITH THE BUILDING CONSTRUCTION RULES. THERE SHALL BE NO DEVIATION FROM THE BUILDING STANDARDS WITHOUT PRIOR WRITTEN APPROVAL FROM THE BUILDING MANAGEMENT.
- WHEN MECHANICAL WORK IS SUBCONTRACTED IT SHALL BE THE MECHANICAL CONTRACTORS RESPONSIBILITY TO COORDINATE SUBCONTRACTORS AND THE ASSOCIATED CONTRACTS. WHEN DISCREPANCIES ARISE PERTAINING TO WHICH CONTRACTOR PROVIDES A PARTICULAR ITEM OF THE MECHANICAL CONTRACT OR WHICH CONTRACTOR PROVIDES FINAL CONNECTIONS FOR A PARTICULAR ITEM OF THE MECHANICAL CONTRACT, IT SHALL BE BROUGHT TO THE ATTENTION OF THE MECHANICAL CONTRACTOR, WHOSE DECISION SHALL BE FINAL.
- ALL MISCELLANEOUS STEEL REQUIRED TO ENSURE PROPER INSTALLATION AND AS SHOWN IN DETAILS FOR PIPING, DUCTWORK, AND EQUIPMENT SHALL BE FURNISHED AND INSTALLED BY THE MECHANICAL CONTRACTOR AND COORDINATED WITH THE GENERAL CONTRACTOR FOR COORDINATION AMONG ALL TRADES.
- PROVIDE ALL MATERIALS, EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
- INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURERS' RECOMMENDATIONS, CONTRACT DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.
- PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 50 FEET OR 50 PIPE DIAMETERS IF ISOLATED EQUIPMENT REFER TO SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL PIPING, DUCTWORK AND EQUIPMENT VIBRATION ISOLATION.
- LOCATE ALL TEMPERATURE, PRESSURE AND FLOW MEASURING DEVICES IN ACCESSIBLE LOCATIONS WITH STRAIGHT SECTION OF PIPE OR DUCT UP- AND DOWNSTREAM AS RECOMMENDED BY THE MANUFACTURER.
- COORDINATE ALL EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS. COORDINATE AND PROVIDE ALL DUCT AND PIPING TRANSITIONS REQUIRED FOR THE FINAL EQUIPMENT CONNECTIONS TO FURNISHED EQUIPMENT. FIELD VERIFY COORDINATE ALL DUCT AND PIPING DIMENSIONS BEFORE FABRICATIONS.
- THE LOCATIONS OF ALL ITEMS SHOWN ON THE DRAWINGS OR CALLED FOR IN THE SPECIFICATIONS THAT ARE NOT DEFINITELY FIXED BY DIMENSIONS ARE APPROXIMATELY ONLY. THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST CONDITIONS AND RESULTS MUST BE DETERMINED BY THE PROJECT SITE CONDITIONS AND SHALL HAVE THE APPROVAL OF THE ENGINEER BEFORE BEING INSTALLED. DO NOT SCALE DRAWINGS.
- PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVICE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT. ACCESS PANELS SHALL BE TURNED OVER TO GENERAL CONTRACTOR INSTALLATION.
- ALL DUCTWORK, PIPING AND EQUIPMENT SUPPORTED FORM STRUCTURAL STEEL SHALL BE COORDINATED WITH GENERAL CONTRACTOR. ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS. PROVIDE BEAM CLAMPS MEETING SMACNA STANDARDS.
- ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC. SHALL BE FIRE STOPPED WITH A DUCT SIMILAR TO 3M OR APPROVED EQUAL, AS SPECIFIED IN THE ARCHITECTURAL DOCUMENTS.
- ALL AIR CONDITIONING CONDENSATE DRAIN LINES FORM EACH AIR HANDLING UNIT AND ROOFTOP UNIT SHALL BE PIPED FULL SIZE OF THE UNIT. DRAIN OUTLET, WITH "P" TRAP, AND PIPED TO NEAREST DRAIN. SEE DETAILS SHOWN ON THE DRAWINGS OR THE CONTRACT SPECIFICATIONS FOR DEPTH OF AIR CONDITIONING CONDENSATE TRAP.
- CERTAIN ITEMS SUCH AS RISERS AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DAMPERS, ETC., ARE INDICATED ON THE CONTRACT DOCUMENT DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS. LOCATIONS OF ALL SUCH ITEMS SHALL BE INDICATED ON SHOP DRAWINGS BY THE INSTALLING CONTRACTOR FOR REVIEW AND APPROVAL DURING THE SHOP DRAWING PROCESS.
- EXACT LOCATIONS OF ALL THERMOSTATS SHALL BE AS PER ARCHITECT/ENGINEER. REFER TO ARCHITECTURAL DWGS FOR THERMOSTAT MOUNTING HEIGHT. MECHANICAL CONTRACTOR SHALL SUBMIT THERMOSTAT LOCATIONS FOR REVIEW PRIOR TO INSTALLATION.
- ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERIOR DIMENSIONS AND DO NOT INCLUDE INCREASED DIMENSION TO ACCOUNT FOR DUCT LINING THICKNESS. REFER TO MECHANICAL SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL DUCT LINING.
- COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS.
- LOCATE ALL MECHANICAL EQUIPMENT FOR UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.
- PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS CONNECTED TO AIR HANDLING UNITS, FANS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AT THE POINT OF CONNECTION TO THE EQUIPMENT UNLESS OTHERWISE INDICATED.
- ALL EQUIPMENT, DUCTWORK & PIPING SHALL BE INSTALLED AT MAXIMUM HEIGHTS ALLOWABLE OR AS REQUIRED TO CLEAR NEW CEILING. DUCTWORK & PIPING SHALL BE OFFSET AS REQUIRED TO AVOID CONFLICTS EVEN IF NOT SPECIFICALLY INDICATED ON PLAN.
- ALL EXPOSED SUPPLY DUCTWORK SHALL HAVE FLANGED AND GASKETTED CONNECTIONS AND SHALL NOT HAVE EXTERNAL INSULATION WRAP. PROVIDE ACOUSTIC LINING AS REQUIRED BY SPECIFICATIONS.
- ALL EXPOSED DUCTWORK SHALL BE FREE OF DEFORMITIES, WRITING, STICKERS OR ANY OTHER MARKINGS. DUCT SURFACES SHALL BE TREATED AS REQUIRED FOR PAINTING. REFER TO ARCHITECTURAL DWGS FOR LOCATIONS OF DUCTWORK TO BE PAINTED.
- ALL DUCTWORK EXPOSED TO MOISTURE SHALL BE ALUMINUM FOR A MIN. OF 10 FEET.
- ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.
- MOUNT ALL SUPPLY REGISTERS AT SAME ELEVATION. ELEVATIONS SHALL BE INDICATED ON THE SHEET/ETAL SHOP DWGS FOR REVIEW BY ARCHITECT/ENGINEER.
- FOR ROOMS WITH EQUIPMENT RACKS, PRODUCTION OF SHEETMETAL SHOP DRAWINGS SHALL INCLUDE ALL COORDINATION REQUIRED TO INDICATE EXACT LOCATIONS OF ALL TECHNICAL EQUIPMENT, EQUIPMENT & LADDER RACK SYSTEMS, ETC. SO AS TO ACCURATELY REVIEW COORDINATED PLACEMENT OF ALL AIR OUTLETS SERVING CRITICAL AREAS.
- PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.
- SMOKE DETECTORS SHALL BE FURNISHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL BE RESPONSIBLE FOR MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.
- MECHANICAL CONTRACTOR TO TAKE PRE-CONSTRUCTION AIR READINGS AT ALL BASE BUILDING MAIN DUCTWORK SERVING SCOPE OF WORK AREA. INCLUDE KEYED PLAN SHOWING READING LOCATIONS AND SUBMIT TO ENGINEER PRIOR TO PRODUCTION OF SHEET METAL SHOP DWGS. REFER TO SPECIFICATIONS FOR MORE INFORMATION.
- AIR BALANCING WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR HIS REPRESENTATIVE AND A COPY OF THE CERTIFIED BALANCING REPORT SHALL BE SUBMITTED TO THE BUILDING MANAGEMENT. PROVIDE 48 HOUR NOTICE TO BUILDING MANAGEMENT FOR THIS WORK.
- MECHANICAL CONTRACTOR TO BALANCE SCOPE OF WORK TO NEW AIR AND WATER QUANTITIES INDICATED UPON PROJECT COMPLETION. BALANCING TO BE PERFORMED BY BASE BUILDING APPROVED CONTRACTOR.
- CONTRACTOR TO PROVIDE TRAVERSE VOLUMETRIC, STATIC PRESSURE AND TEMPERATURE READINGS AT ALL BASE BUILDING MAIN SUPPLY, RETURN AND EXHAUST SHAFTS. READINGS TO BE INCLUDED IN FINAL BALANCING REPORT.
- REFER TO MECHANICAL PIPING PLAN FOR THERMOSTAT LOCATIONS AND THERMOSTAT CONTROL PLAN.

MKDA

902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

ARCHITECT OF RECORD:

JOHN F. GIUSEFFI, ARCHITECT, PC.

902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

snowflake

7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:

AMA CONSULTING ENGINEERS

825 8TH AVE #18
NEW YORK, NY 10019
212.944.7722

AVIT/SEC CONSULTANT:

NORTHLAND CONTROLS

44150 S GRIMMER BLVD
FREMONT, CA 94538

FOOD SERVICE DESIGN CONSULTANT:

COLBURN GUYETTE

100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

DESIGN CONSULTANT:

JPC ARCHITECTS

909 112th Ave NE, Suite 206
Bellevue, WA 98004

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7 TIMES SQUARE, NEW YORK, NY		
FLOORS 29, 30, 31		
DRAWING TITLE:		
LEVEL 30 - MECHANICAL DUCTWORK PLAN		
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[illegible]

- ① DUCTWORK TO BE CONSTRUCTED OF 16-GAUGE OUTER DUCT W/1" ACOUSTICAL LINING, TIED LINER AND STANDARD GAUGE PERFORATED INNER DUCT TO MATCH BUILDING STANDARD FOR FAN DISCHARGE DUCTWORK.
- ② EXPOSED DUCTWORK SHALL BE OVAL OR ROUND SPIRAL DUCT CONSTRUCTED OF GALVANIZED STEEL WITH ACOUSTICAL LINING AS PER SPECIFICATIONS, FABRICATE AND INSTALL ALL FLEX SMOCA'S STANDARD, WITH ALL SEAMS, TIGHT AND CONNECTIONS SUITABLE FOR PRESSURE CLASS, SUPPORT AND HANG DUCT PER CODE REQUIREMENTS.
- ③ UNIT LOCATION TO BE COORDINATED WITH IDY LAYOUT, LOCATION AND POSITIONING SHALL ALLOW FOR CLEARANCES AS SPECIFIED BY MANUFACTURER. IF HANGING, UNIT SHOULD BE VIBRATION INSULATED AS DESCRIBED FOR MECHANICAL EQUIPMENT IN SPECIFICATIONS.

1. ALL WORK SHALL CONFORM TO THE LATEST BUILDING STANDARDS. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO OBTAIN A COPY OF THE BUILDING DEPARTMENT STREET VIEWING REQUIREMENTS FOR ORDER TO BECOME TOTALLY FAMILIAR WITH THE BUILDING CONSTRUCTION RULES. THERE SHALL BE NO DEVIATION FROM THE BUILDING STANDARDS WITHOUT PRIOR WRITTEN APPROVAL FROM THE BUILDING MANAGEMENT.

2. WHEN MECHANICAL WORK IS SUBCONTRACTED IT SHALL BE THE MECHANICAL CONTRACTOR'S RESPONSIBILITY TO COORDINATE SUBCONTRACTORS AND THE ASSOCIATED CONTRACTS. WHEN DISCREPANCIES ARISE PERTAINING TO WHICH CONTRACT PROVIDES A PARTICULAR ITEM OF THE MECHANICAL CONTRACT OR WHEN THE CONTRACTS PROVIDE DIFFERENT CONNECTIONS FOR A PARTICULAR ITEM OF THE MECHANICAL CONTRACT, IT SHALL BE BROUGHT TO THE ATTENTION OF THE MECHANICAL CONTRACTOR, WHOSE DECISION SHALL BE FINAL.

3. ALL MISCELLANEOUS STEEL REQUIRED TO ENSURE PROPER INSTALLATION AND AS SHOWN IN DETAILS FOR PIPING, DUCTWORK, AND EQUIPMENT SHALL BE FURNISHED AND WELDED BY THE MECHANICAL CONTRACTOR AND COORDINATED WITH THE GENERAL CONTRACTOR FOR COORDINATION AMONG ALL TRADES.

4. PROVIDE ALL MATERIALS, EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.

5. INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURERS' RECOMMENDATIONS, CONTRACT DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.

6. PROVIDE VIBRATION ISOLATION FOR ALL MECHANICAL EQUIPMENT TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.

7. PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN FREE FIRE EXITS. PROVIDE VIBRATION ISOLATION FOR ALL EQUIPMENT TO MEET SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL PIPING, DUCTWORK AND EQUIPMENT VIBRATION ISOLATION.

8. LOCATE ALL TEMPERATURE, PRESSURE AND FLOW MEASURING DEVICES IN ACCESSIBLE LOCATIONS WITH STRAIGHT SECTION OF PIPE OR DUCT UP AND DOWNSTREAM AS RECOMMENDED BY THE MANUFACTURER.

9. COORDINATE ALL EQUIPMENT CONNECTIONS WITH MANUFACTURER'S CERTIFIED DRAWINGS. COORDINATE AND PROVIDE ALL DUCT AND PIPING TRAFFICS REQUIRED FOR THE EQUIPMENT CONNECTIONS TO BE FURNISHED EQUIPMENT. FIELD VERIFY COORDINATE ALL DUCT AND PIPING DIMENSIONS BEFORE FABRICATIONS.

10. THE LOCATIONS OF ALL ITEMS SHOWN ON THE DRAWINGS OR CALLED FOR IN THE SPECIFICATIONS THAT ARE NOT DEFINITELY FIXED BY DIMENSIONS ARE APPROXIMATELY ONLY. THE EXACT LOCATIONS NECESSARY TO SECURE THE BEST AND MOST EFFICIENT RESULTS MUST BE DETERMINED BY THE INSTALLER. THE CONTRACTOR SHALL HAVE THE APPROVAL OF THE ENGINEER BEFORE BEING INSTALLED. DO NOT SCALE DRAWINGS.

11. PROVIDE ACCESS PANELS FOR INSTALLATION IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVICE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER MECHANICAL EQUIPMENT. ACCESS PANELS SHALL BE TURNED OVER TO GENERAL CONTRACTOR INSTALLATION.

12. ALL DUCTWORK, PIPING AND EQUIPMENT SUPPORTED FORM STRUCTURAL STEEL SHALL BE COORDINATED WITH GENERAL CONTRACTOR, ALL ATTACHMENTS TO STEEL BAR JOISTS, TRUSSES, OR JOIST GIRDERS SHALL BE AT PANEL POINTS. PROVIDE BEAM CLIPS, MEETING SMOKE STANDARDS.

13. ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC. SHALL BE FIRE RESISTANT AND APPROXIMATELY TO 3M OR APPROVED EQUAL, AS SPECIFIED IN THE ARCHITECTURAL DOCUMENTS.

14. ALL AIR CONDITIONING CONDENSATE DRAIN LINES FOR EACH AIR HANDLING UNIT AND ROOFTOP UNIT SHALL BE PIPED FULL SIZE OF THE UNIT DRAIN OUTLET, WITH "P" TRAP AND PIPED TO NEAREST DRAIN. SEE DETAILS SHOWN ON THE DRAWINGS OR THE CONTRACT SPECIFICATIONS FOR DEPTH OF AIR CONDITIONING CONDENSATE TRAP.

15. CERTAIN ITEMS SUCH AS RISERS AND DROPS IN DUCTWORK, ACCESS DOORS, VOLUME DUCTS, AND INSULATION SHALL BE SHOWN ON THE DRAWINGS. PROVIDE DRAWINGS FOR CLARITY FOR A SPECIFIC LOCATION REQUIREMENT AND SHALL NOT BE INTERPRETED AS THE EXTENT OF THE REQUIREMENTS FOR THESE ITEMS. LOCATIONS OF ALL SUCH ITEMS SHALL BE INDICATED BY THE CONTRACTOR BY THE INSTALLING CONTRACTOR FOR REVIEW AND APPROVAL DURING THE SHOP DRAWING PROCESS.

16. EXACT LOCATIONS OF ALL THERMOSTATS SHALL BE AS PER ARCHITECT/ENGINEER. REFER TO ARCHITECTURAL DWGS FOR THERMOSTAT MOUNTING HEIGHT. MECHANICAL CONTRACTOR SHALL SUBMIT THERMOSTAT LOCATIONS FOR REVIEW PRIOR TO INSTALLATION.

17. ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERIOR DIMENSIONS. THE INSULATION SHALL BE ADDED TO THE DIMENSIONS TO ACCOMMODATE DUCT LINING THICKNESS. REFER TO MECHANICAL SPECIFICATIONS FOR EXACT REQUIREMENTS OF ALL DUCT LINING.

18. COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS, LIGHTING, AND OTHER CEILING ITEMS.

19. LOCATE ALL MECHANICAL EQUIPMENT FOR UNOBSTRUCTED ACCESS TO UNIT ACCESS PANELS, CONTROLS AND VALVING.

20. PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS CONNECTED TO AIR HANDLING UNITS, FANS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AT THE POINT OF CONNECTION TO FIRE EXISTING EQUIPMENT UNLESS OTHERWISE INDICATED.

21. ALL EQUIPMENT, DUCTWORK & PIPING SHALL BE INSTALLED AT MAXIMUM HEIGHTS ALLOWABLE OR AS REQUIRED TO CLEAR NEW CEILING. DUCTWORK & PIPING SHALL BE OFFSET AS REQUIRED TO AVOID CONFLICTS EVEN IF NOT SPECIFICALLY INDICATED ON PLAN.

22. ALL EXPOSED SUPPLY DUCTWORK SHALL HAVE PLANGED AND GASKETTED CONNECTIONS AND SHALL NOT HAVE EXTERNAL INSULATION WRAP. PROVIDE ACCESS LINING AS REQUIRED FOR DECONTAMINATION.

23. ALL EXPOSED DUCTWORK SHALL BE FREE OF DEFORNATIONS, WRITING, STICKERS OR ANY OTHER READING LOCATIONS AND SUBMIT TO ENGINEER PRIOR TO PRODUCTION OF SHEET METAL SHOP DWGS. REFER TO SPECIFICATIONS FOR LOCATIONS OF DUCTWORK TO BE PAINTED.

24. ALL DUCTWORK EXPOSED TO MOISTURE SHALL BE ALUMINUM FOR A MIN. OF 10 FEET.

25. ALL SUPPLY DUCTWORK SHOWN TO HAVE ROUND DIMENSIONS SHALL BE SPIRAL ROUND DUCTWORK.

26. MOUNT ALL SUPPLY REGISTERS AT SAME ELEVATION. ELEVATIONS SHALL BE INDICATED ON THE SHEETMETAL SHOP DWGS FOR REVIEW BY ARCHITECT/ENGINEER.

27. FOR ROOMS WITH EQUIPMENT RACKS: PRODUCTION OF SHEETMETAL SHOP DRAWINGS SHALL INCLUDE ALL COORDINATION REQUIRED TO INDICATE EXACT LOCATIONS FOR ALL MECHANICAL EQUIPMENT, EQUIPMENT A LOADER RACK SYSTEMS, AIR HANDLING EQUIPMENT, AND REVIEW COORDINATED PLACEMENT OF ALL AIR OUTLETS SERVING CRITICAL AREAS.

28. PROVIDE ACCESS DOORS IN DUCTWORK TO PROVIDE ACCESS FOR ALL SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, VOLUME DAMPERS, HUMIDIFIERS, COILS, AND OTHER ITEMS LOCATED IN THE DUCTWORK WHICH REQUIRE SERVICE AND/OR INSPECTION.

29. SMOKE DETECTORS SHALL BE NICHED AND WIRED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL PROVIDE THE ACCESS TO THE MOUNTING THE SMOKE DETECTOR IN DUCTWORK AS SHOWN ON THE DRAWINGS AND IN ACCORDANCE WITH MANUFACTURERS PRINTED INSTRUCTIONS.

30. MECHANICAL CONTRACTOR TO TAKE PRE-CONSTRUCTION AIR READINGS AT ALL BASE BUILDING MAIN DUCTWORK SERVING SCOPE OF WORK AREA. INCLUDE KEYPED PLAN AND EXISTING AIR READINGS. SUBMIT TO ENGINEER PRIOR TO PRODUCTION OF SHEET METAL SHOP DWGS. REFER TO SPECIFICATIONS FOR MORE INFORMATION.

31. AIR BALANCING WORK SHALL BE WITNESSED BY THE BUILDING'S CHIEF ENGINEER OR HIS REPRESENTATIVE AND A COPY OF THE CERTIFIED BALANCING REPORT SHALL BE SUBMITTED TO THE BUILDING MANAGEMENT. PROVIDE 48 HOUR NOTICE TO BUILDING MANAGEMENT FOR THIS WORK.

32. MECHANICAL CONTRACTOR TO BALANCE SCOPE OF WORK TO NEW AIR AND WATER QUANTITIES INDICATED UPON PROJECT COMPLETION. BALANCING TO BE PERFORMED BY A BASE BUILDING APPROVED CONTRACTOR.

33. CONTRACTOR TO PROVIDE TRAVERSE VOLUMETRIC, STATIC PRESSURE AND TEMPERATURE READINGS AT ALL BASE BUILDING MAIN SUPPLY, RETURN AND EXHAUST SHAFTS. READINGS TO BE INCLUDED IN FINAL BALANCING REPORT.

34. REFER TO MECHANICAL PIPING PLAN FOR THERMOSTAT LOCATIONS AND THERMOSTAT CONTROL PLAN.

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	05/22/2026	ISSUE FOR FILING
1	04/22/2026	ADDENDUM #1
	04/03/2026	ISSUE FOR PRICING
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PROJECT:

SNOWFLAKE

7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

LEVEL 31 - MECHANICAL DUCTWORK PLAN

SEAL & SIGNATURE

DATE: 05.22.2026

SCALE: As indicated

PROJECT NO. _____

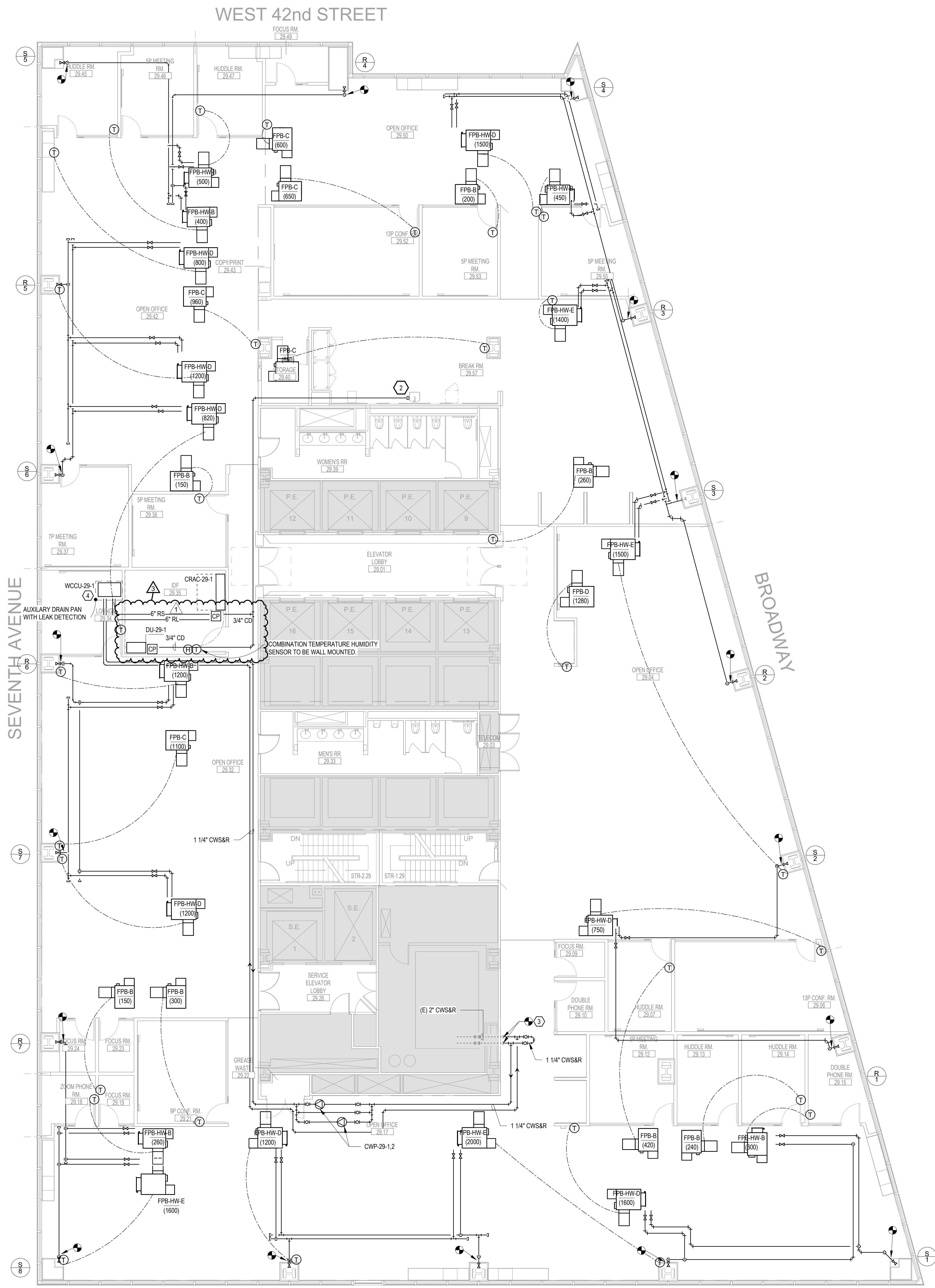
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- KEY NOTES**
- MECHANICAL CONTRACTOR TO COORDINATE WITH PLUMBING CONTRACTOR TO PROVIDE COLD WATER TO CRAC UNIT (INTERNAL HUMIDIFIER) IN IDF ROOM.
 - PROVIDE FUNNEL DRAIN FOR CONDENSATE DRAIN UNDER SINK. REFER TO PLUMBING DRAWINGS FOR FUNNEL DRAIN LOCATION.
 - CONNECT TO EXISTING CONDENSER WATER VALVES. CONTRACTOR TO VERIFY SUPPLY AND RETURN CONNECTIONS IN FIELD. PROVIDE PRESSURE MEASUREMENTS AT SUPPLY AND RETURN CONNECTIONS WITH BUILDING PUMPS AT FULL FLOW AND SUBMIT WITH TENANT PUMPS.
 - PROVIDE ALL LEAK DETECTION PER BUILDING RULES AND REGULATIONS. CONNECT ALL LEAK DETECTION TO SHUT ASSOCIATED ISOLATION VALVE AND ALARM THE BUILDING BMS. COORDINATE WITH PLUMBING CONTRACTOR. SEE MECHANICAL SPECIFICATION.

- PIPING NOTES**
- PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PIPING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
 - PROVIDE AN AIR VENT AT THE HIGH POINT OF EACH DROP IN THE HEATING WATER, AND OTHER CLOSED WATER PIPING. ALL PIPING SHALL GRADE TO LOW POINTS. PROVIDE HOSE END DRAINING VALVES AT THE BOTTOM OF ALL RISERS AND LOW POINTS.
 - UNLESS OTHERWISE NOTED, ALL PIPING IS OVERHEAD, TIGHT TO UNDERSIDE OF STRUCTURE OF SLAB, WITH SPACE FOR INSULATION OF REQUIRED.
 - INSTALL PIPING SO THAT ALL VALVES, STRAINERS, UNIONS, TRAPS, FLANGES, AND OTHER APPURTENANCES REQUIRING ACCESS ARE ACCESSIBLE.
 - ALL VALVES SHALL BE INSTALLED SO THAT VALVE REMAINS IN SERVICE WHEN EQUIPMENT OR PIPING ON EQUIPMENT SIDE OF VALVE IS REMOVED.
 - ALL BALANCING VALVES AND BUTTERFLY VALVES SHALL BE PROVIDED WITH POSITION INDICATORS AND MAXIMUM ADJUSTABLE STOPS (MEMORY STOPS).
 - ALL VALVES (EXCEPT CONTROL VALVES) AND STRAINERS SHALL BE FULL SIZE OF PIPE BEFORE REDUCING SIZE TO MAKE CONNECTIONS TO EQUIPMENT AND CONTROLS.
 - UNIONS AND/OR FLANGES SHALL BE INSTALLED AT EACH PIECE OF EQUIPMENT, IN BYPASSES, AND IN LONG PIPING RUNS (100 FEET OR MORE) TO PERMIT DISASSEMBLY FOR ALTERATION AND REPAIRS.
 - ALL PIPING SHALL CLEAR DOORS AND WINDOWS.
 - ALL VALVES SHALL BE ADJUSTED FOR SMOOTH AND EASY OPERATION.
 - ALL PIPING WORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFF SETS IN PIPING AROUND OBSTRUCTIONS SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.
 - PROVIDE FLEXIBLE CONNECTIONS IN ALL PIPING SYSTEMS CONNECTED TO PUMPS, CHILLERS, COOLING TOWERS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION EXCEPT WATER COILS. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AS CLOSE TO THE EQUIPMENT AS POSSIBLE OR AS INDICATED ON THE DRAWINGS.
 - PROVIDE LINE SIZE STRAINER UPSTREAM OF EACH AUTOMATIC VALVE. PROVIDE SHUTOFF VALVE ON EACH SIDE OF STRAINER.
 - PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 30 FEET IF ISOLATED EQUIPMENT (EXCEPT AT BASE ELBOW SUPPORTS AND ANCHOR POINTS) THROUGHOUT MECHANICAL EQUIPMENT ROOM.
 - REFRIGERANT LINES TO BE SIZED PER MANUFACTURER RECOMMENDATION.



902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

ARCHITECT OF RECORD:

JOHN F. GIUSEFFI, ARCHITECT, PC.

902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

snowflake

7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:

AMA CONSULTING ENGINEERS

825 8TH AVE #18
NEW YORK, NY 10019
212.944.7722

AVIT/SEC CONSULTANT:

NORTHLAND CONTROLS

44150 S GRIMMER BLVD
FREMONT, CA 94538

FOOD SERVICE DESIGN CONSULTANT:

COLBURN GUYETTE

100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

DESIGN CONSULTANT:

JPC ARCHITECTS

909 112th Ave NE, Suite 206
Bellevue, WA 98004

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PROJECT:

SNOWFLAKE

7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

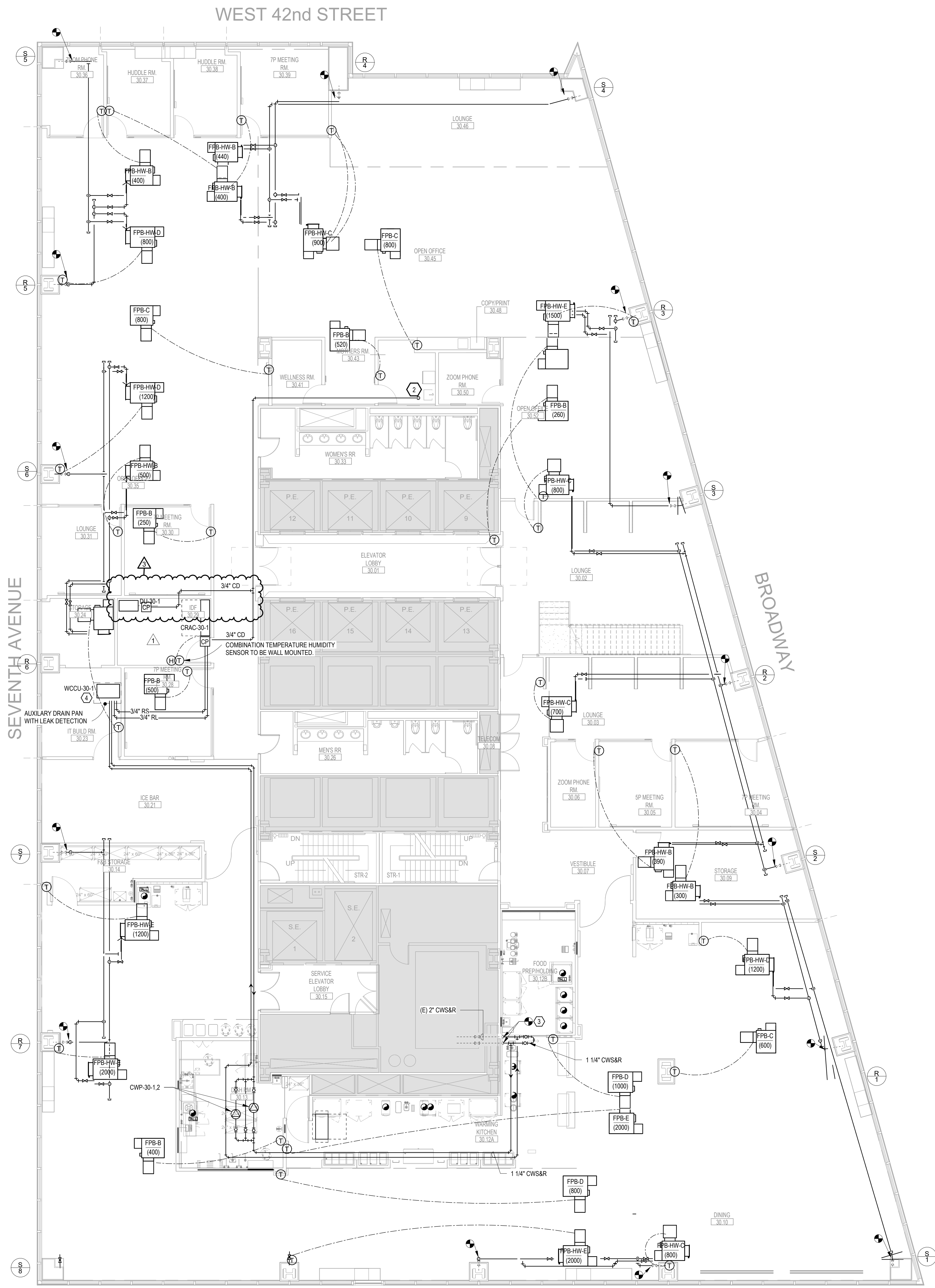
LEVEL 29 - MECHANICAL PIPING PLAN

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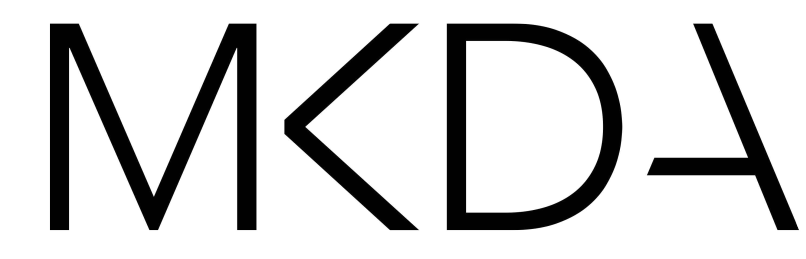
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- ### KEY NOTES
- MECHANICAL CONTRACTOR TO COORDINATE WITH PLUMBING CONTRACTOR TO PROVIDE COLD WATER TO CRAC UNIT (INTERNAL HUMIDIFIER) IN IDF ROOM.
 - PROVIDE FUNNEL DRAIN FOR CONDENSATE DRAIN UNDER SINK. REFER TO PLUMBING DRAWINGS FOR FUNNEL DRAIN LOCATION.
 - CONNECT TO EXISTING CONDENSER WATER VALVES. CONTRACTOR TO VERIFY SUPPLY AND RETURN CONNECTIONS IN FIELD. PROVIDE PRESSURE MEASUREMENTS AT SUPPLY AND RETURN CONNECTIONS WITH BUILDING PUMPS AT FULL FLOW AND SUBMIT WITH TENANT PUMPS.
 - PROVIDE ALL LEAK DETECTION PER BUILDING RULES AND REGULATIONS. CONNECT ALL LEAK DETECTION TO SHUT ASSOCIATED ISOLATION VALVE AND ALARM THE BUILDING BMS. COORDINATE WITH PLUMBING CONTRACTOR. SEE MECHANICAL SPECIFICATION.

- ### PIPING NOTES
- PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PIPING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
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 - ALL BALANCING VALVES AND BUTTERFLY VALVES SHALL BE PROVIDED WITH POSITION INDICATORS AND MAXIMUM ADJUSTABLE STOPS (MEMORY STOPS).
 - ALL VALVES (EXCEPT CONTROL VALVES) AND STRAINERS SHALL BE FULL SIZE OF PIPE BEFORE REDUCING SIZE TO MAKE CONNECTIONS TO EQUIPMENT AND CONTROLS.
 - UNIONS AND/OR FLANGES SHALL BE INSTALLED AT EACH PIECE OF EQUIPMENT, IN BYPASSES, AND IN LONG PIPING RUNS (100 FEET OF MORE) TO PERMIT DISASSEMBLY FOR ALTERATION AND REPAIRS.
 - ALL PIPING SHALL CLEAR DOORS AND WINDOWS.
 - ALL VALVES SHALL BE ADJUSTED FOR SMOOTH AND EASY OPERATION.
 - ALL PIPING WORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFF SETS IN PIPING AROUND OBSTRUCTIONS SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.
 - PROVIDE FLEXIBLE CONNECTIONS IN ALL PIPING SYSTEMS CONNECTED TO PUMPS, CHILLERS, COOLING TOWERS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION EXCEPT WATER COOLS. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AS CLOSE TO THE EQUIPMENT AS POSSIBLE OR AS INDICATED ON THE DRAWINGS.
 - PROVIDE LINE SIZE STRAINER UPSTREAM OF EACH AUTOMATIC VALVE. PROVIDE SHUTOFF VALVE ON EACH SIDE OF STRAINER.
 - PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 50 FEET IF ISOLATED EQUIPMENT (EXCEPT AT BASE ELBOW SUPPORTS AND ANCHOR POINTS) THROUGHOUT MECHANICAL EQUIPMENT ROOM.
 - REFRIGERANT LINES TO BE SIZED PER MANUFACTURER RECOMMENDATION.



902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

ARCHITECT OF RECORD:
JOHN F. GIUSEFFI, ARCHITECT, PC.
902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:
AMA CONSULTING ENGINEERS
825 8TH AVE #18
NEW YORK, NY 10019
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FOOD SERVICE DESIGN CONSULTANT:
COLBURN GUYETTE
100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

DESIGN CONSULTANT:
JPC ARCHITECTS
909 112th Ave NE, Suite 206
Bellevue, WA 98004

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1	04/22/2026	ADDENDUM #1
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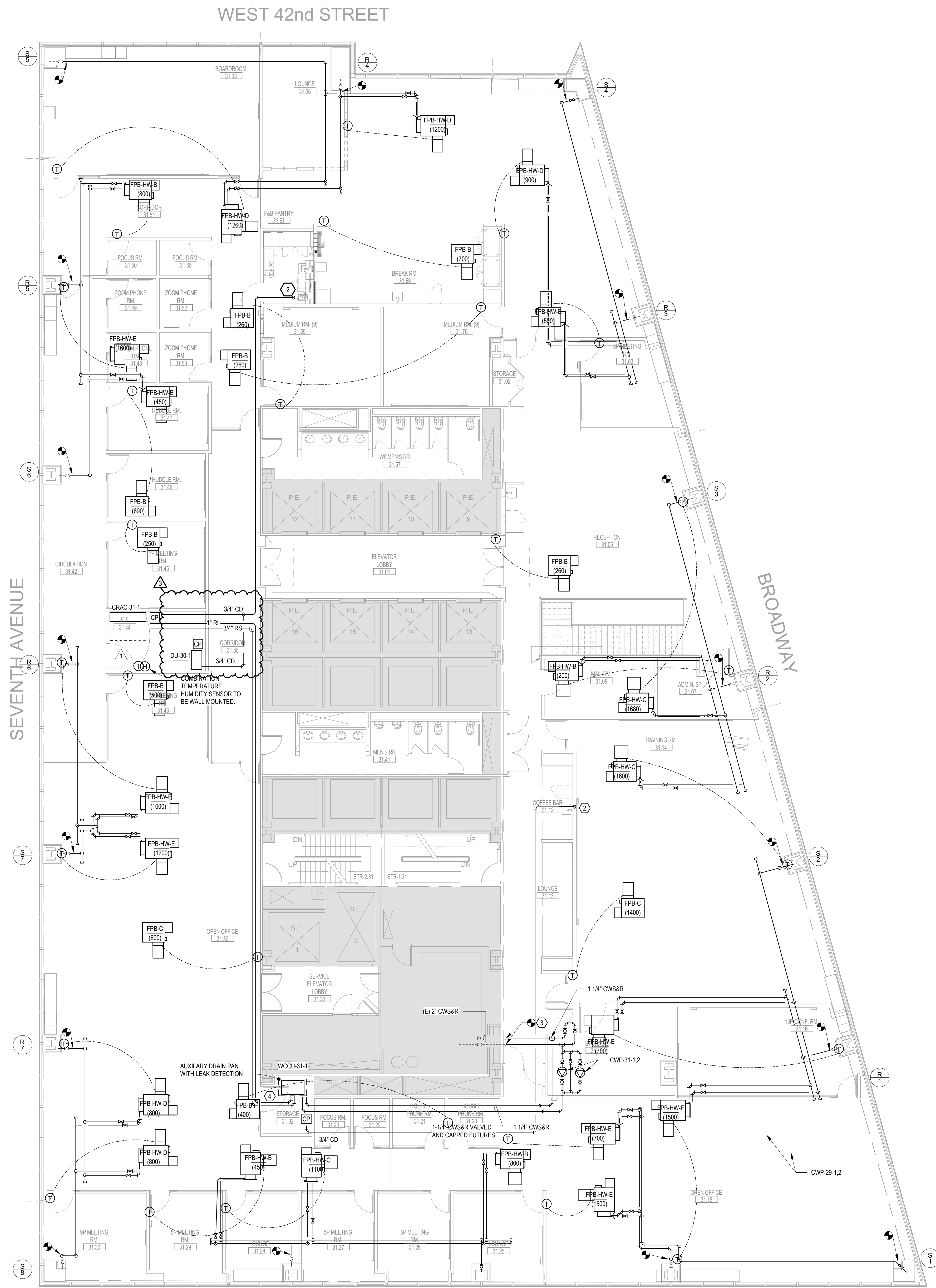
PROJECT:
SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:
LEVEL 30 - MECHANICAL PIPING PLAN

SEAL & SIGNATURE	DATE: 05.22.2026
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NYC DOB NUMBER
M01397257-S1



KEY NOTES

- MECHANICAL CONTRACTOR TO COORDINATE WITH PLUMBING CONTRACTOR TO PROVIDE COLD WATER TO CRAC UNIT (INTERNAL HUMIDIFIER) IN IDF ROOM.
- PROVIDE FUNNEL DRAIN FOR CONDENSATE DRAIN UNDER SINK. REFER TO PLUMBING DRAWINGS FOR FUNNEL DRAIN LOCATION.
- CONNECT TO EXISTING CONDENSER WATER VALVES. CONTRACTOR TO VERIFY SUPPLY AND RETURN CONNECTIONS IN FIELD. PROVIDE PRESSURE MEASUREMENTS AT SUPPLY AND RETURN CONNECTIONS WITH BUILDING PUMPS AT FULL FLOW AND SUBMIT WITH TENANT PUMPS.
- PROVIDE ALL LEAK DETECTION PER BUILDING RULES AND REGULATIONS. CONNECT ALL LEAK DETECTION TO SHUT ASSOCIATED ISOLATION VALVE AND ALARM THE BUILDING BMS. COORDINATE WITH PLUMBING CONTRACTOR. SEE MECHANICAL SPECIFICATION.

PIPING NOTES

- PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE PIPING SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
- PROVIDE AN AIR VENT AT THE HIGH POINT OF EACH DROP IN THE HEATING WATER, AND OTHER CLOSED WATER PIPING. ALL PIPING SHALL GRADE TO LOW POINTS. PROVIDE HOSE END DRAINING VALVES AT THE BOTTOM OF ALL RISERS AND LOW POINTS.
- UNLESS OTHERWISE NOTED, ALL PIPING IS OVERHEAD, TIGHT TO UNDERSIDE OF STRUCTURE OF SLAB, WITH SPACE FOR INSULATION OF REQUIRED.
- INSTALL PIPING SO THAT ALL VALVES, STRAINERS, UNIONS, TRAPS, FLANGES, AND OTHER APPURTENANCES REQUIRING ACCESS ARE ACCESSIBLE.
- ALL VALVES SHALL BE INSTALLED SO THAT VALVE REMAINS IN SERVICE WHEN EQUIPMENT OR PIPING ON EQUIPMENT SIDE OF VALVE IS REMOVED.
- ALL BALANCING VALVES AND BUTTERFLY VALVES SHALL BE PROVIDED WITH POSITION INDICATORS AND MAXIMUM ADJUSTABLE STOPS (MEMORY STOPS).
- ALL VALVES (EXCEPT CONTROL VALVES) AND STRAINERS SHALL BE FULL SIZE OF PIPE BEFORE REDUCING SIZE TO MAKE CONNECTIONS TO EQUIPMENT AND CONTROLS.
- UNIONS AND/OR FLANGES SHALL BE INSTALLED AT EACH PIECE OF EQUIPMENT, IN BYPASSES, AND IN LONG PIPING RUNS (100 FEET OF MORE) TO PERMIT DISASSEMBLY FOR ALTERATION AND REPAIRS.
- ALL PIPING SHALL CLEAR DOORS AND WINDOWS.
- ALL VALVES SHALL BE ADJUSTED FOR SMOOTH AND EASY OPERATION.
- ALL PIPING WORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFF SETS IN PIPING AROUND OBSTRUCTIONS SHALL BE PROVIDED AT NO ADDITIONAL COST TO THE OWNER.
- PROVIDE FLEXIBLE CONNECTIONS IN ALL PIPING SYSTEMS CONNECTED TO PUMPS, CHILLERS, COOLING TOWERS, AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION EXCEPT WATER COOLS. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AS CLOSE TO THE EQUIPMENT AS POSSIBLE OR AS INDICATED ON THE DRAWINGS.
- PROVIDE LINE SIZE STRAINER UPSTREAM OF EACH AUTOMATIC VALVE. PROVIDE SHUTOFF VALVE ON EACH SIDE OF STRAINER.
- PROVIDE VIBRATION ISOLATORS FOR ALL PIPING SUPPORTS CONNECTED TO AND WITHIN 50 FEET IF ISOLATED EQUIPMENT (EXCEPT AT BASE ELBOW SUPPORTS AND ANCHOR POINTS) THROUGHOUT MECHANICAL EQUIPMENT ROOM.
- REFRIGERANT LINES TO BE SIZED PER MANUFACTURER RECOMMENDATION.

MKDA

902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

ARCHITECT OF RECORD:

JOHN F. GIUSEFFI, ARCHITECT, PC.
902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

snowflake
7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:

AMA CONSULTING ENGINEERS

825 8TH AVE #18
NEW YORK, NY 10019
212.944.7722

AVIT/SEC CONSULTANT:

NORTHLAND CONTROLS

44150 S GRIMMER BLVD
FREMONT, CA 94538

FOOD SERVICE DESIGN CONSULTANT:

COLBURN GUYETTE

100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

DESIGN CONSULTANT:

JPC ARCHITECTS

909 112th Ave NE Suite 206
Bellevue, WA 98004

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SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

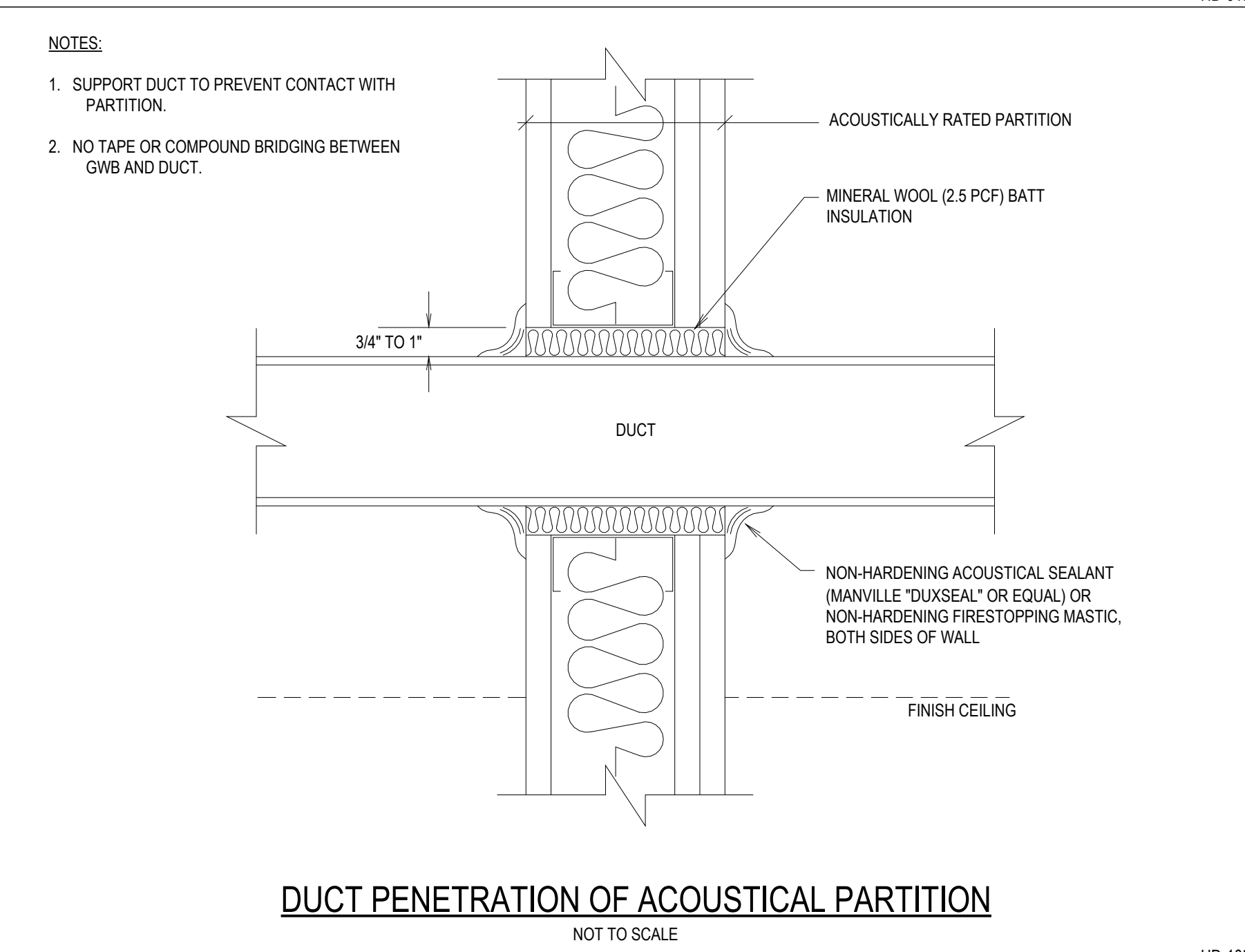
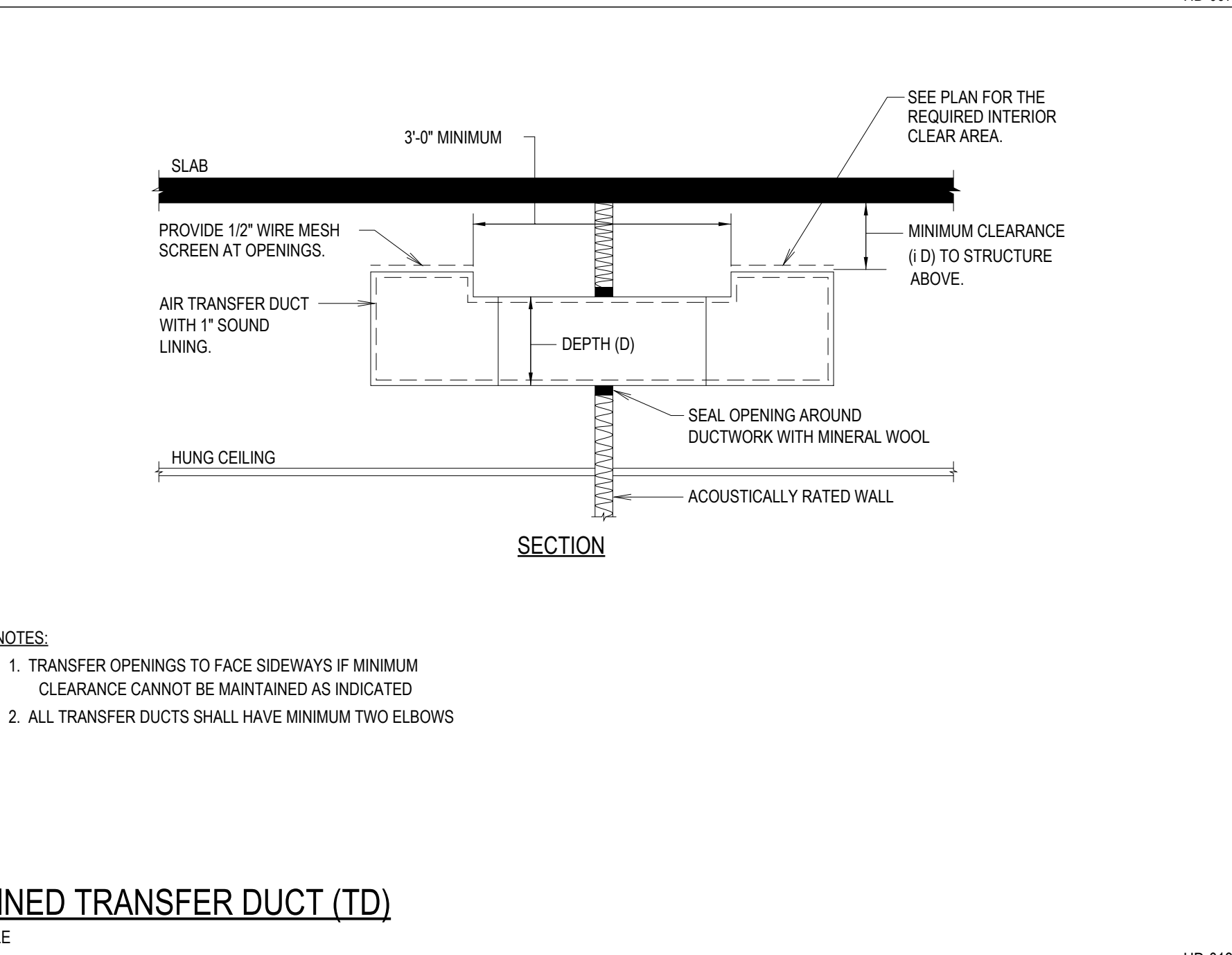
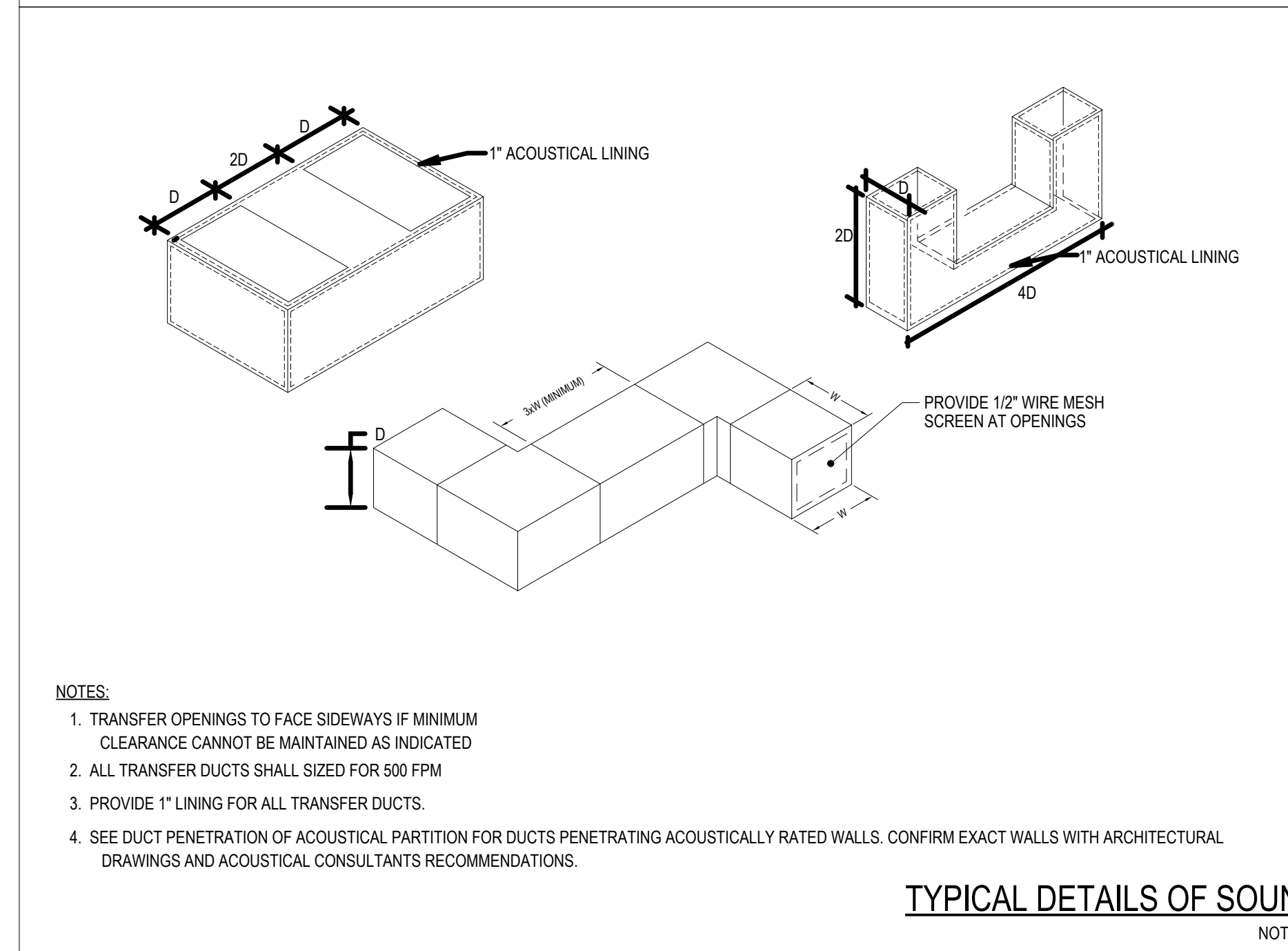
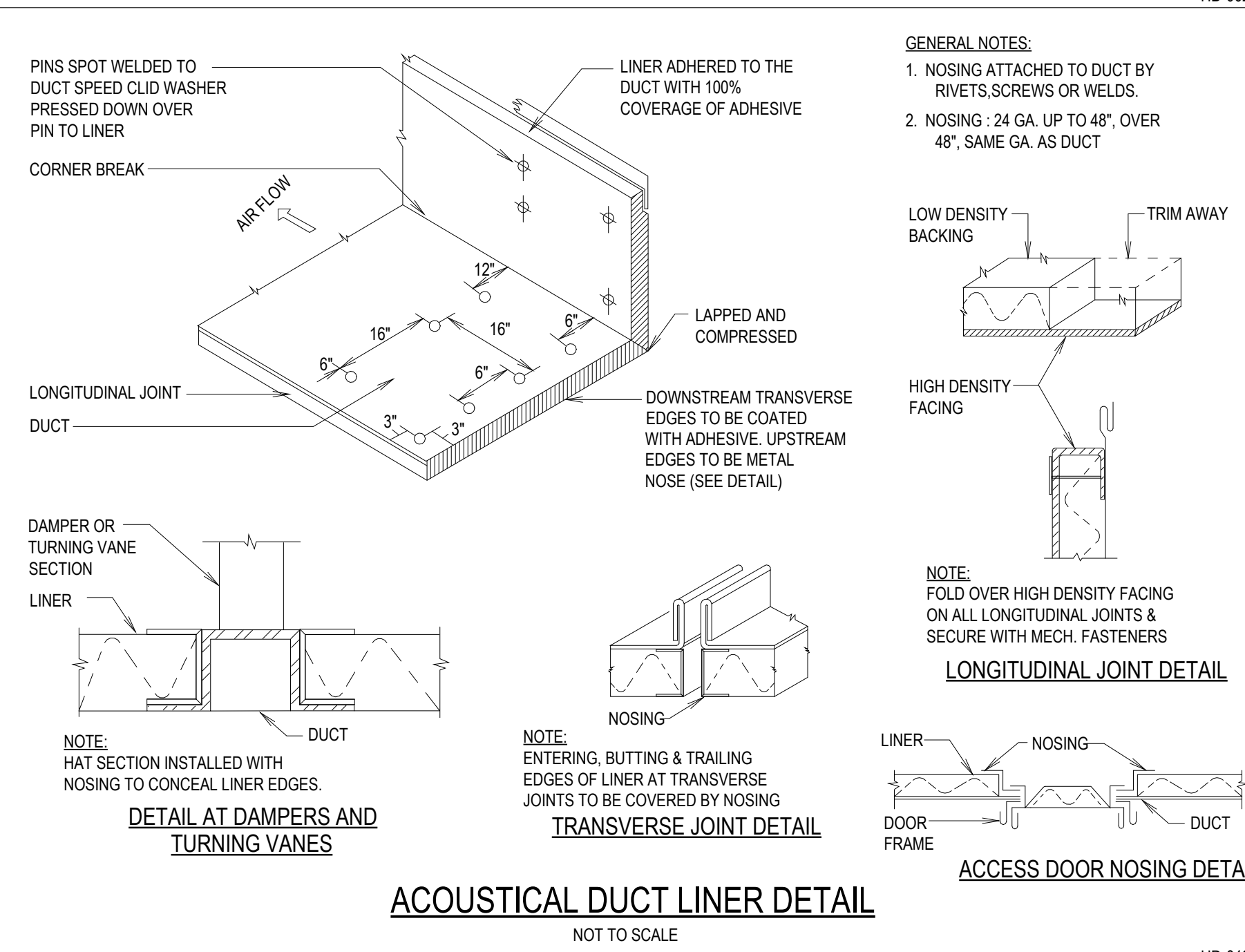
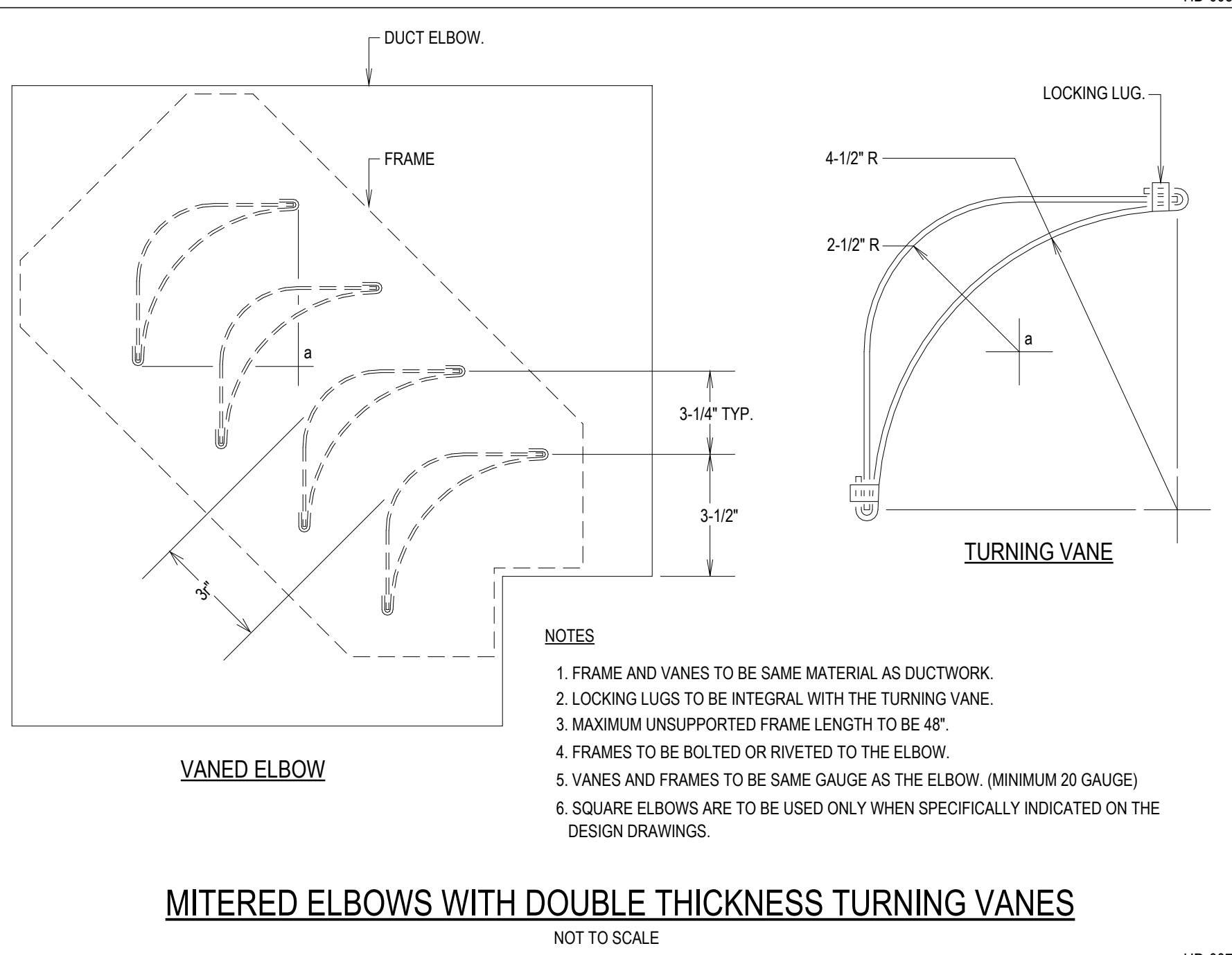
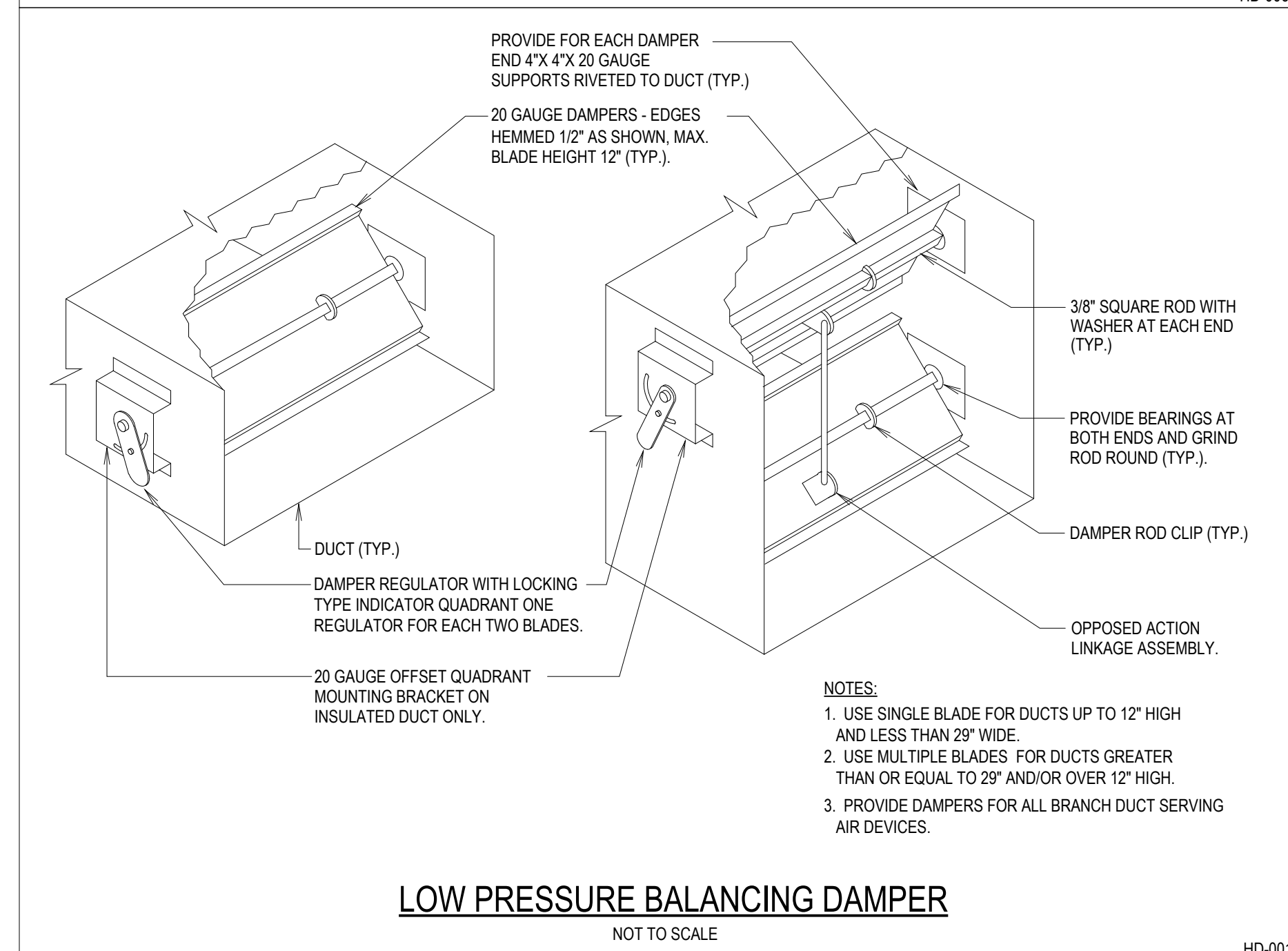
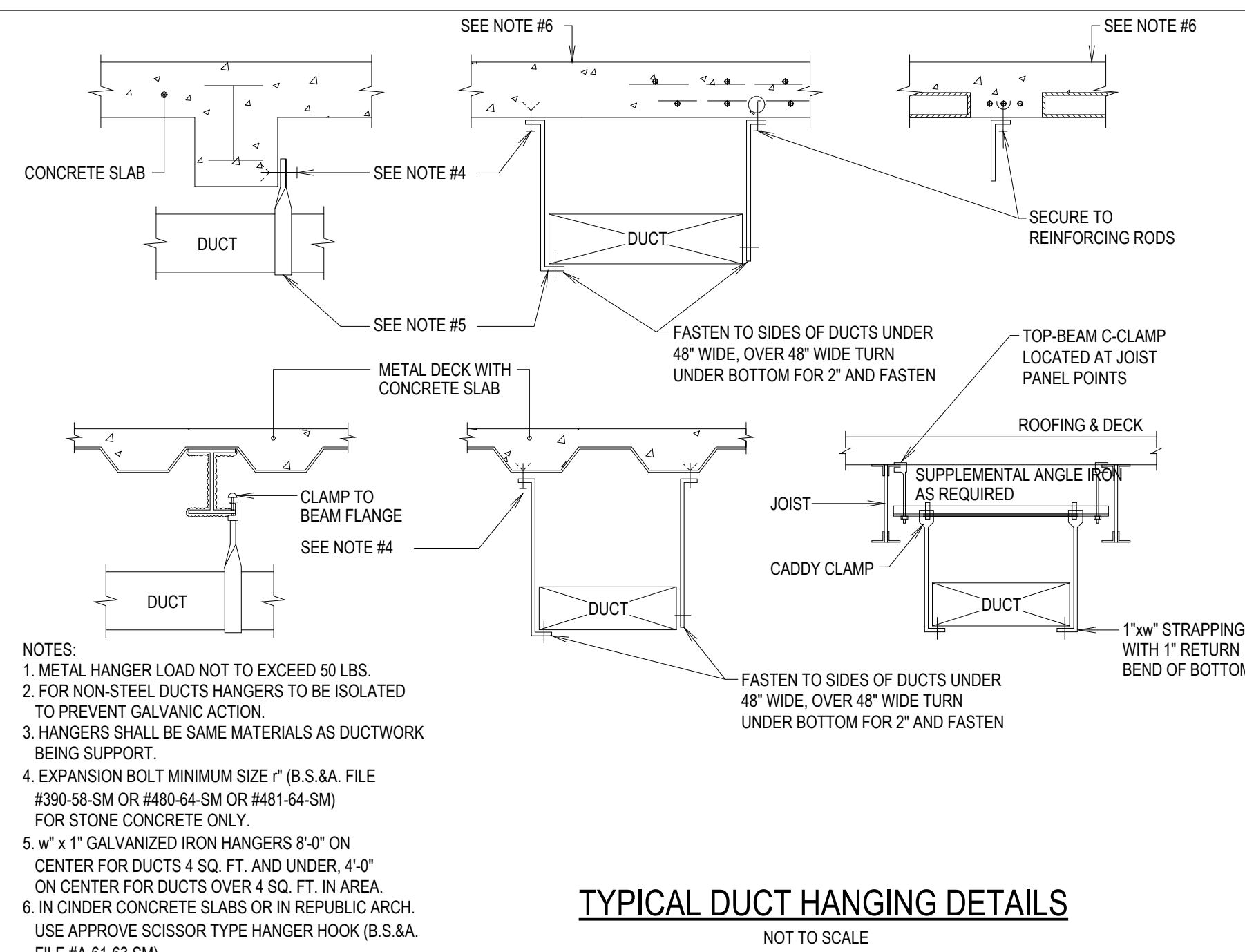
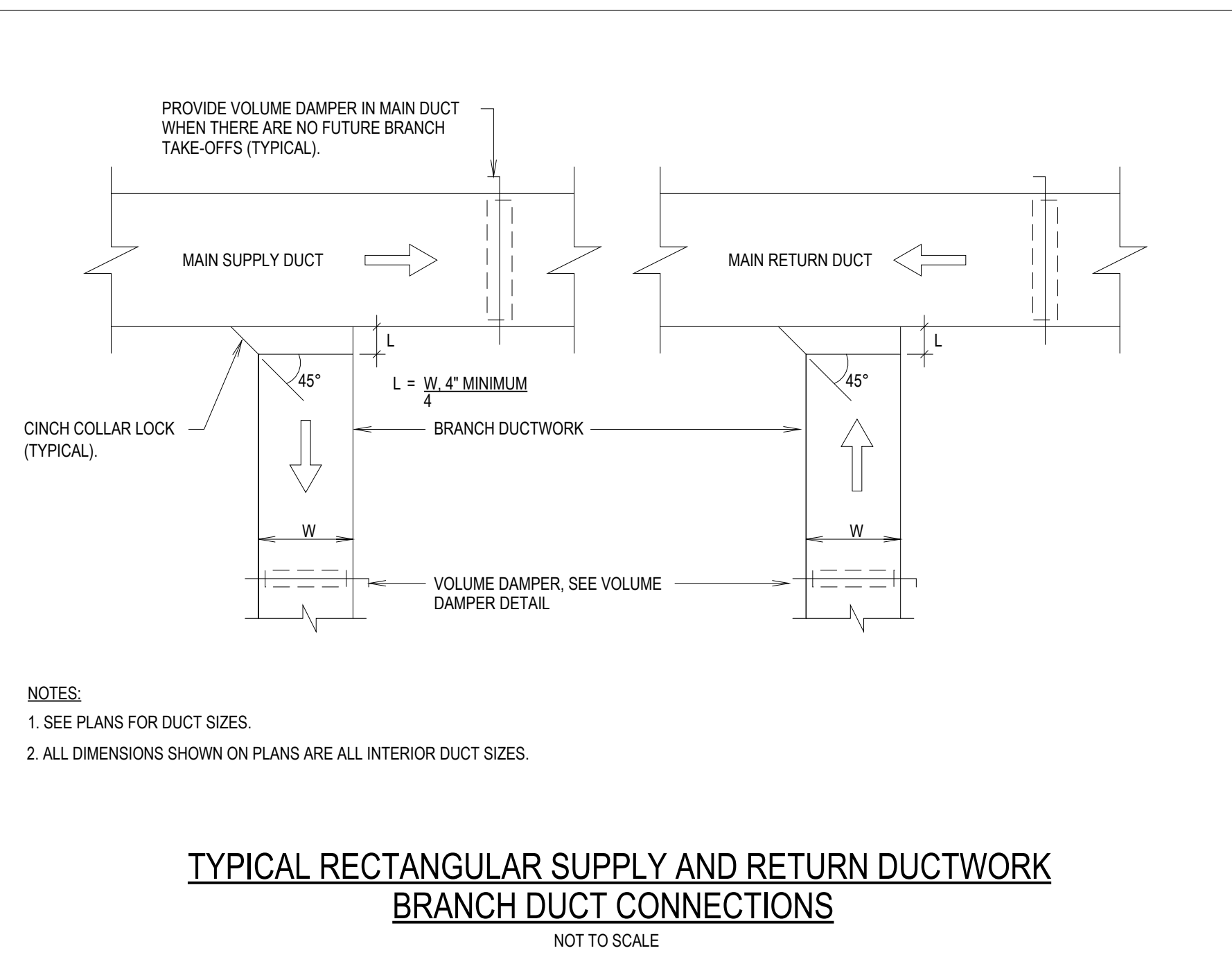
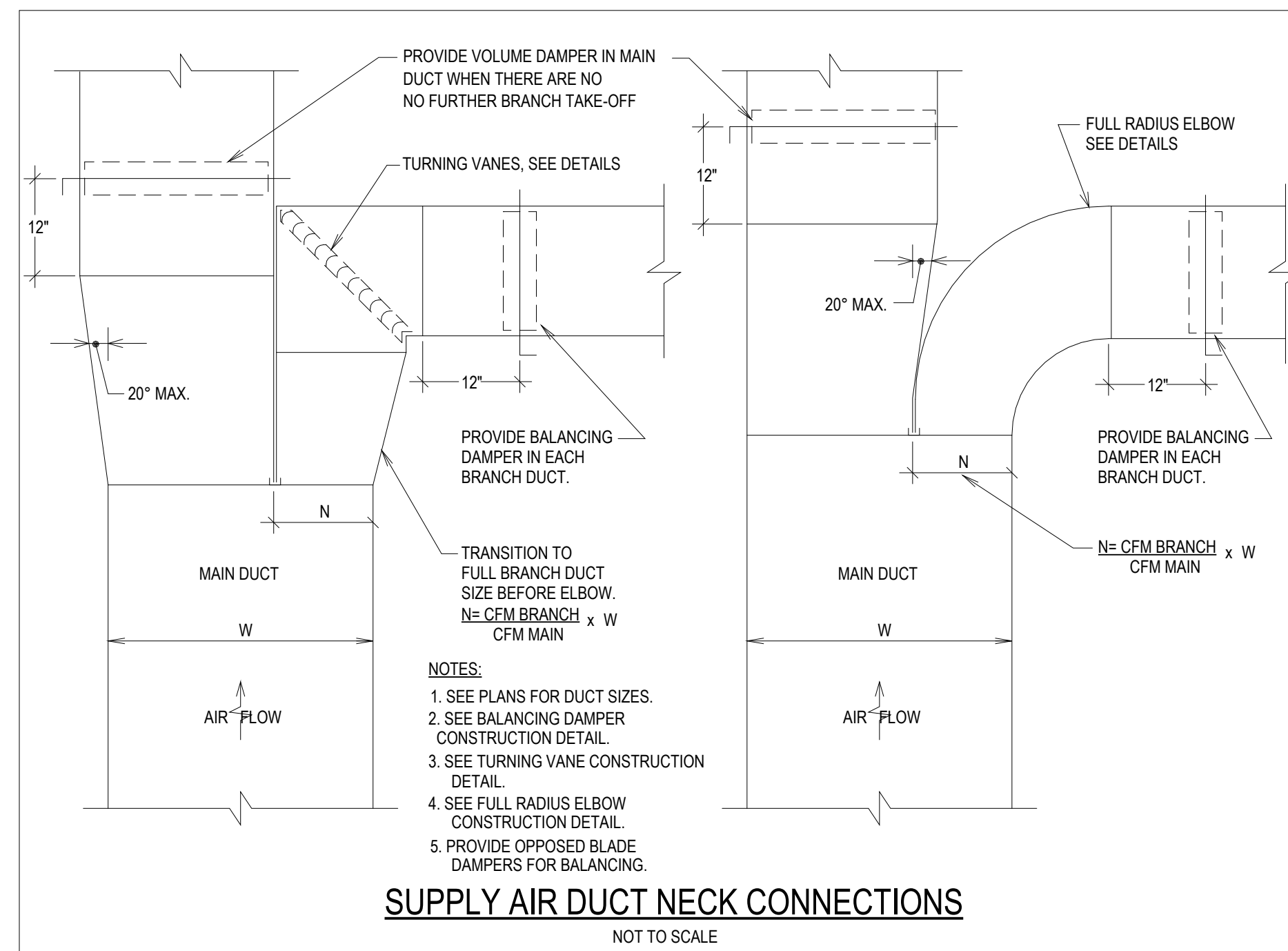
LEVEL 31 - MECHANICAL PIPING PLAN

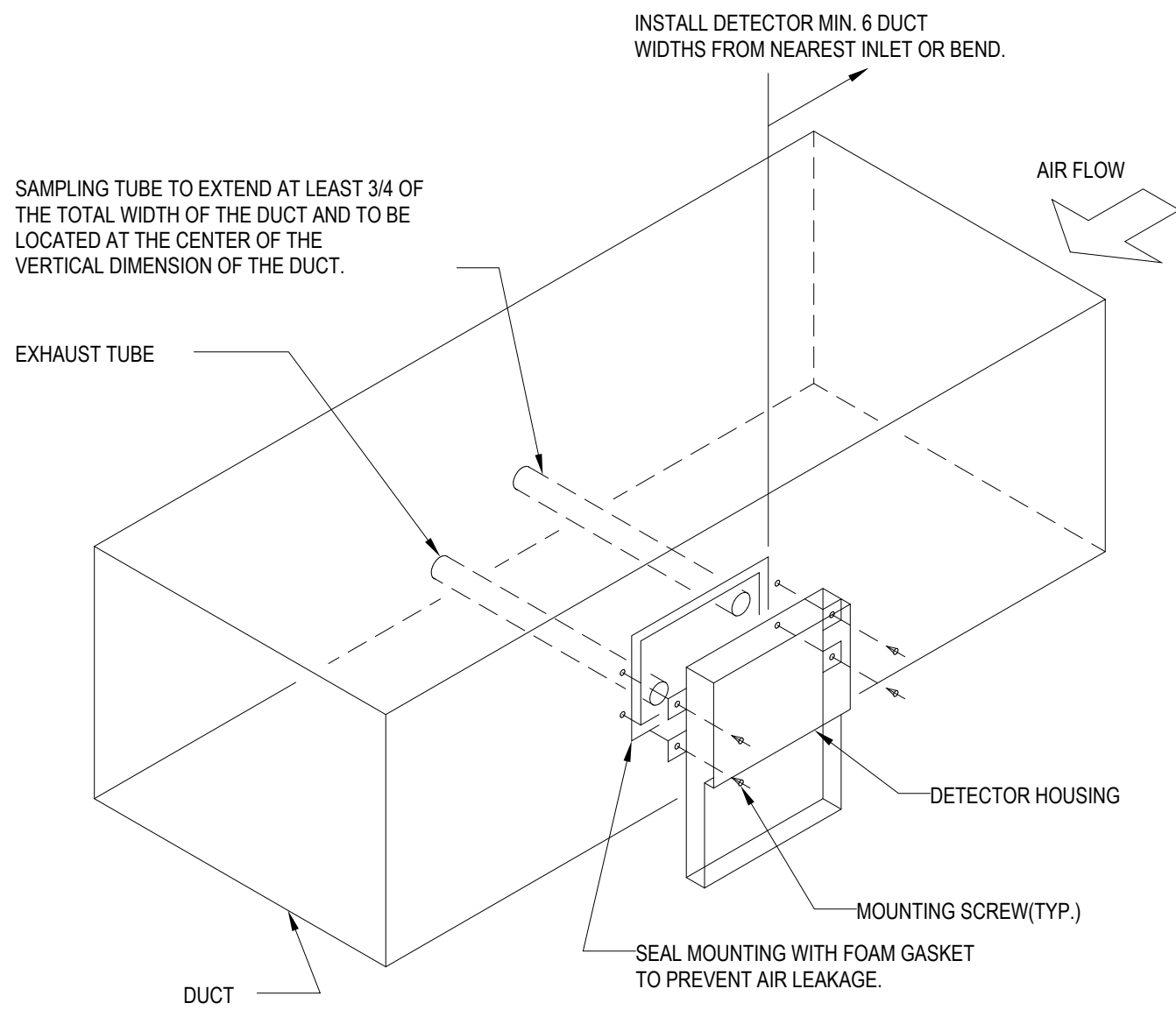
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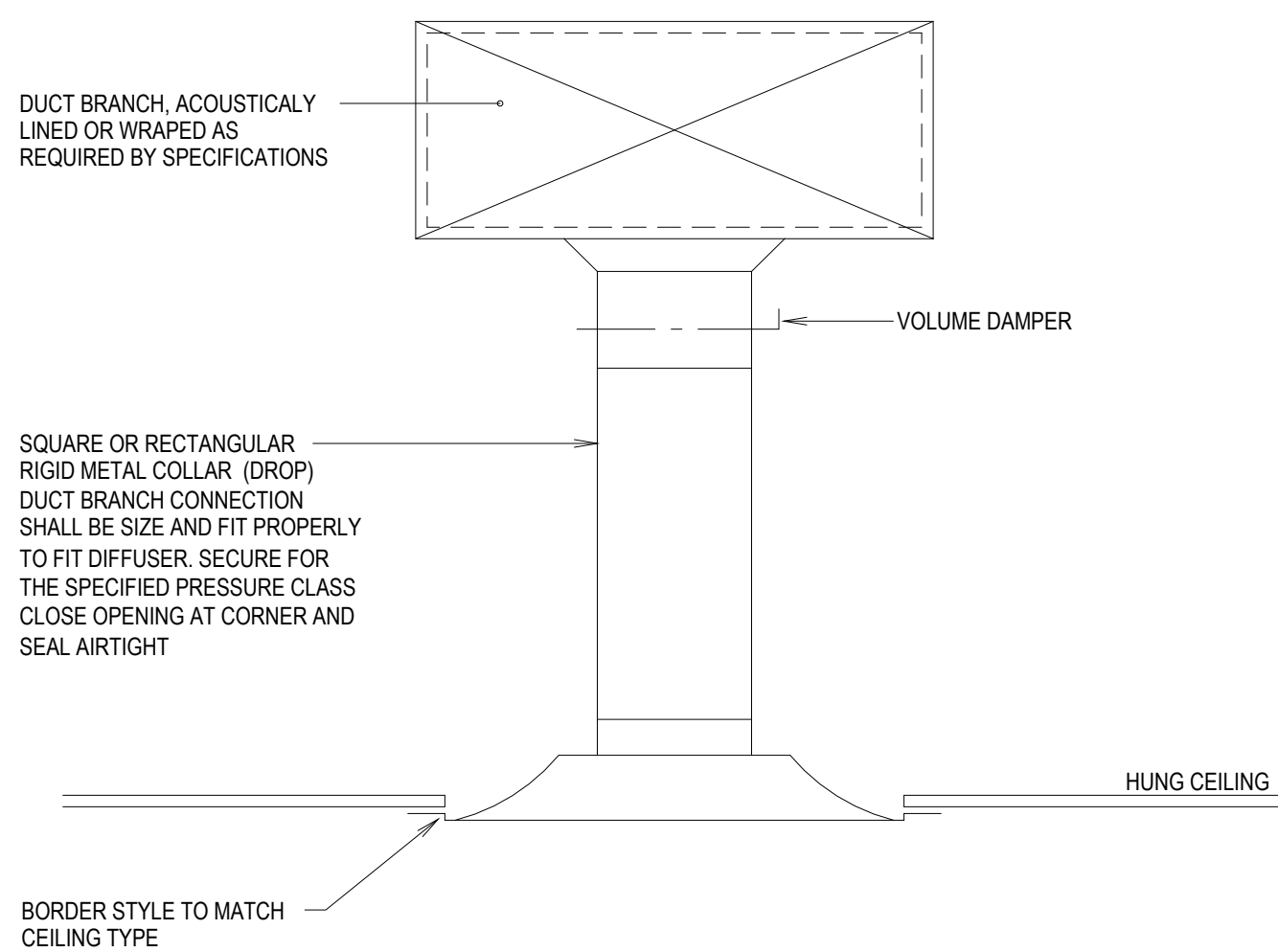




DUCT SMOKE DETECTOR MOUNTING DETAIL

NOT TO SCALE

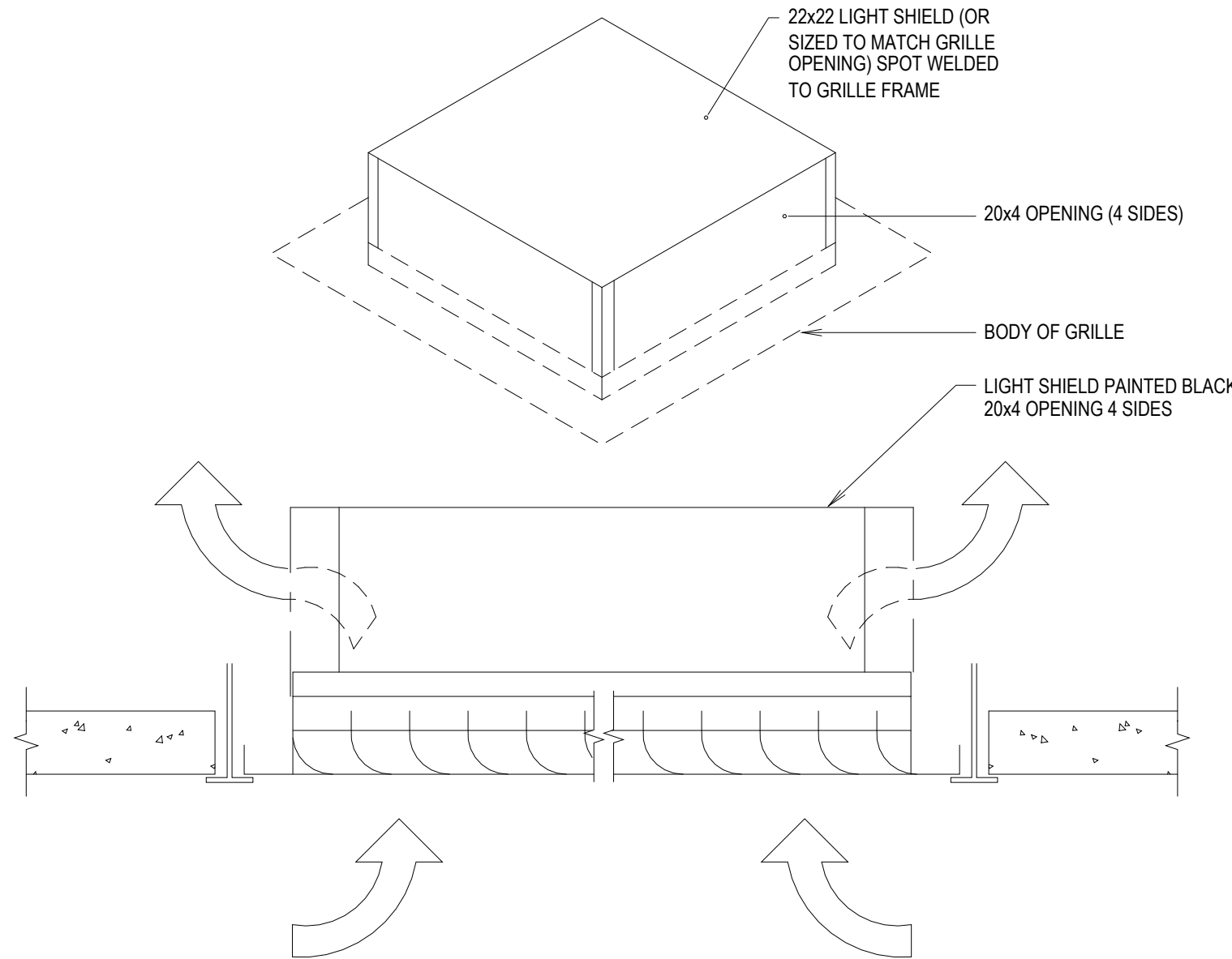
HD-114



TYPICAL RIGID DIFFUSER CONNECTION TO DUCTWORK

NOT TO SCALE

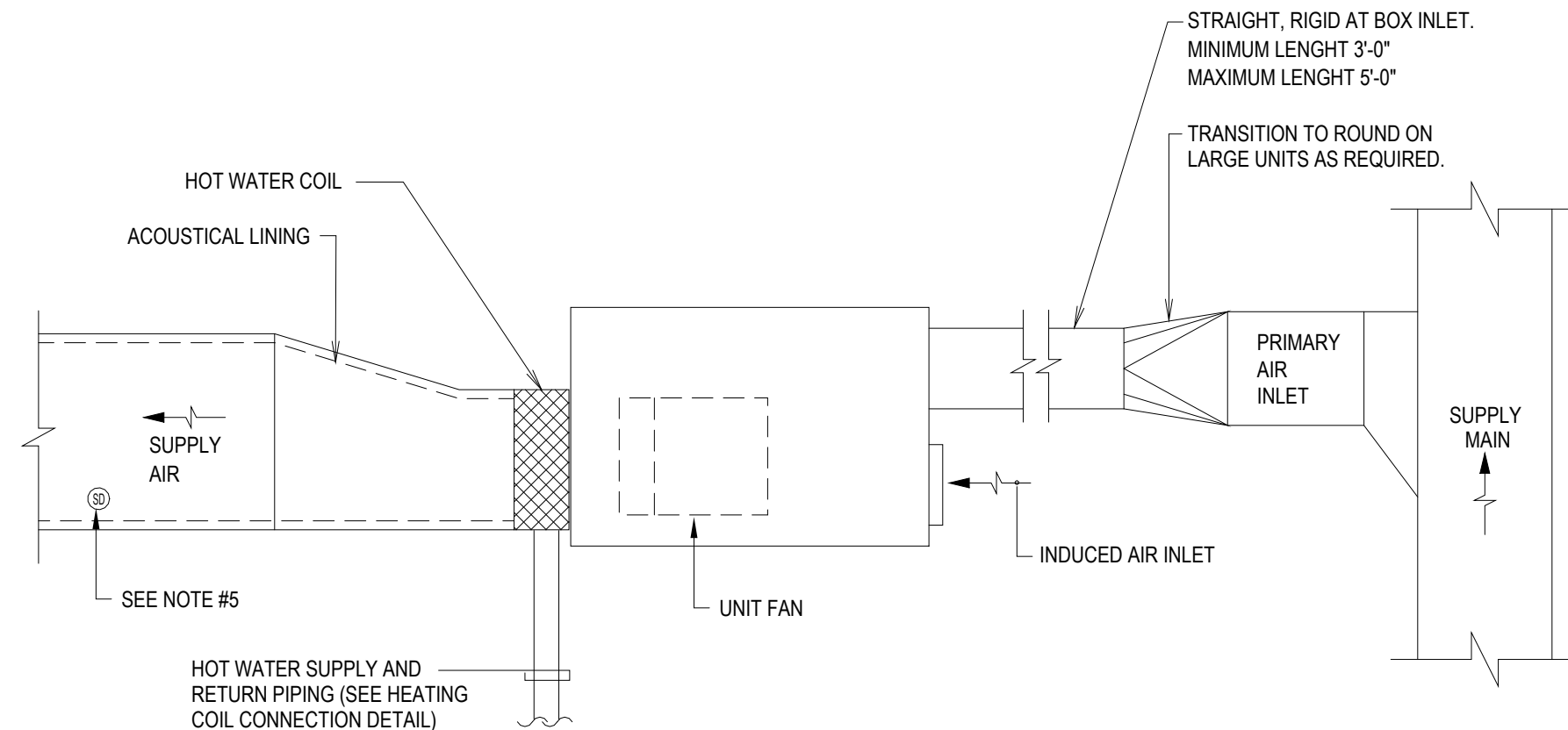
HD-095



RETURN GRILLE WITH LIGHT SHIELD DETAIL

NOT TO SCALE

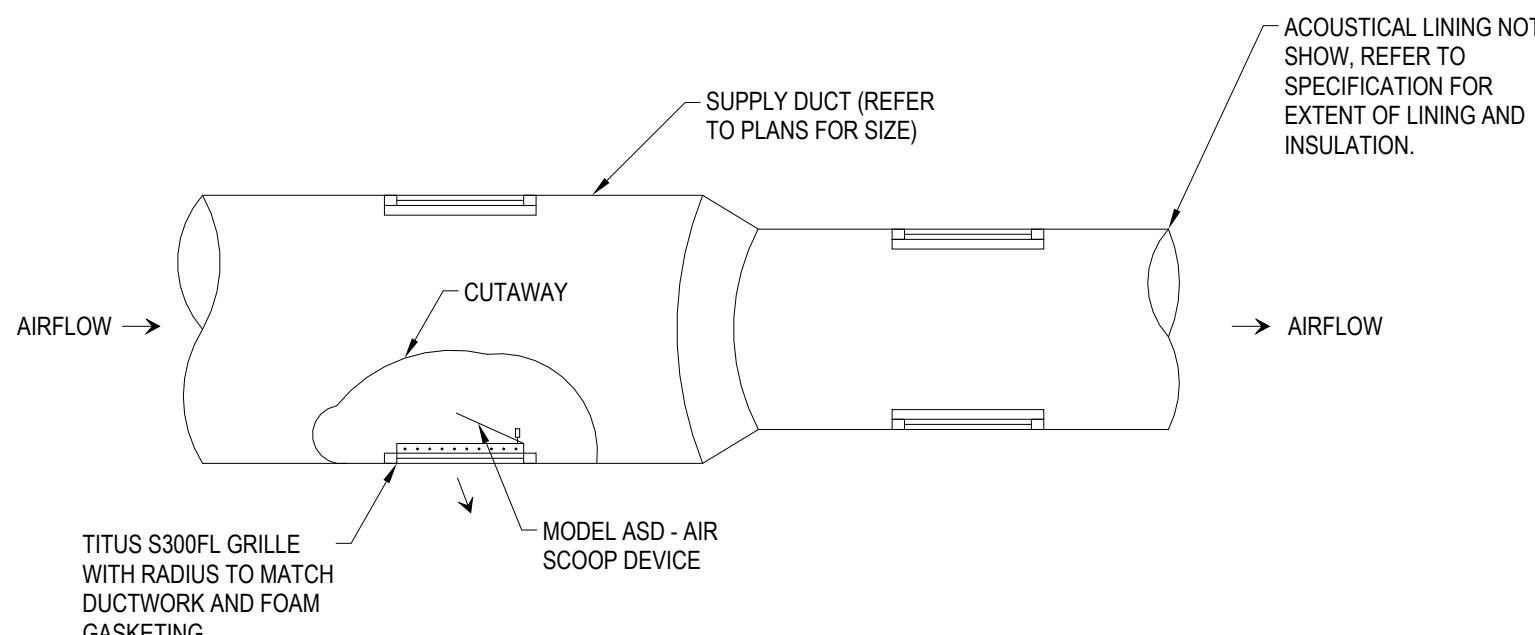
HD-082



FAN POWERED TERMINAL UNIT DETAIL WITH HOT WATER COIL

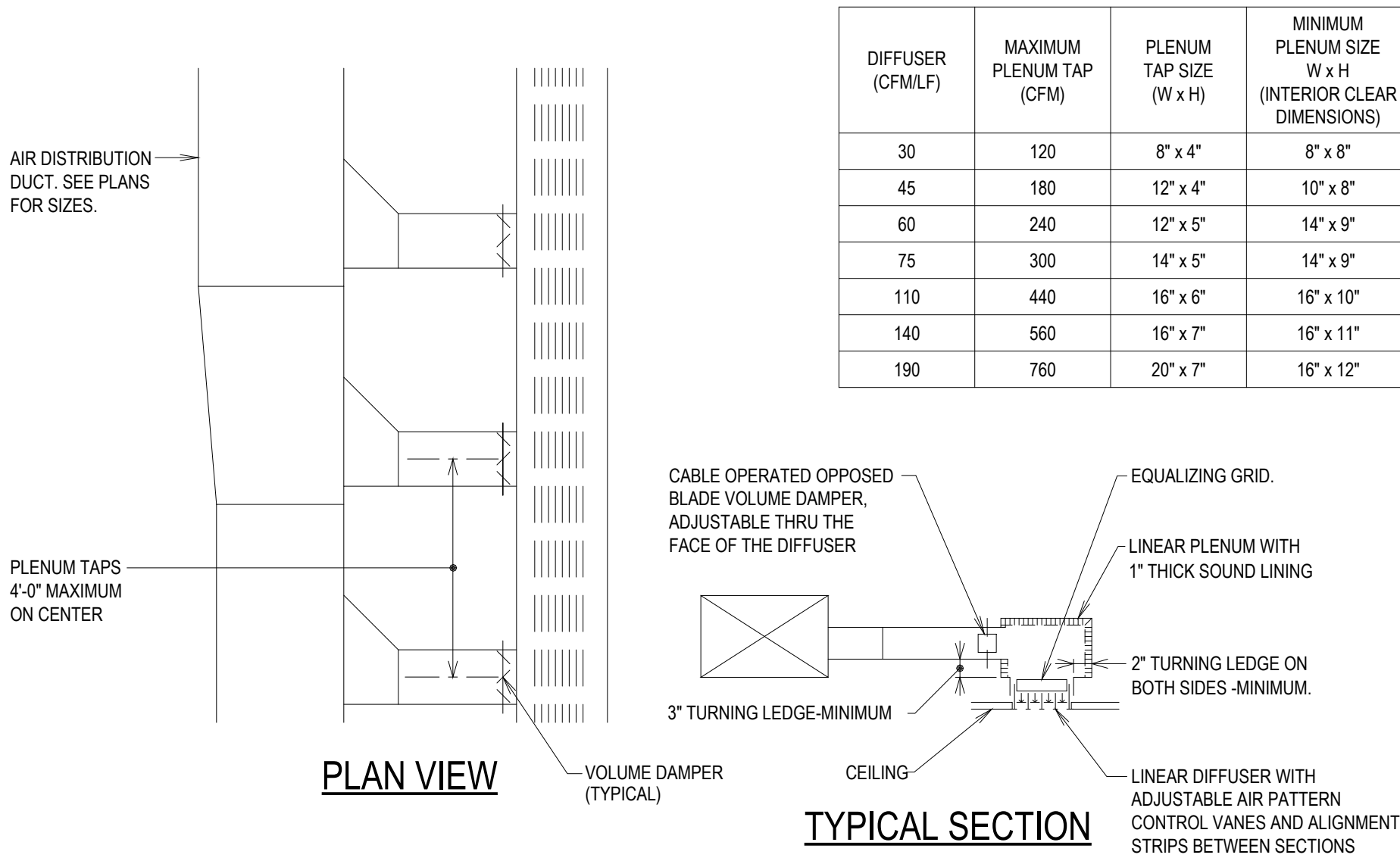
NOT TO SCALE

HD-014



EXPOSED FLAT-OVAL DUCT SUPPLY GRILLE DETAIL

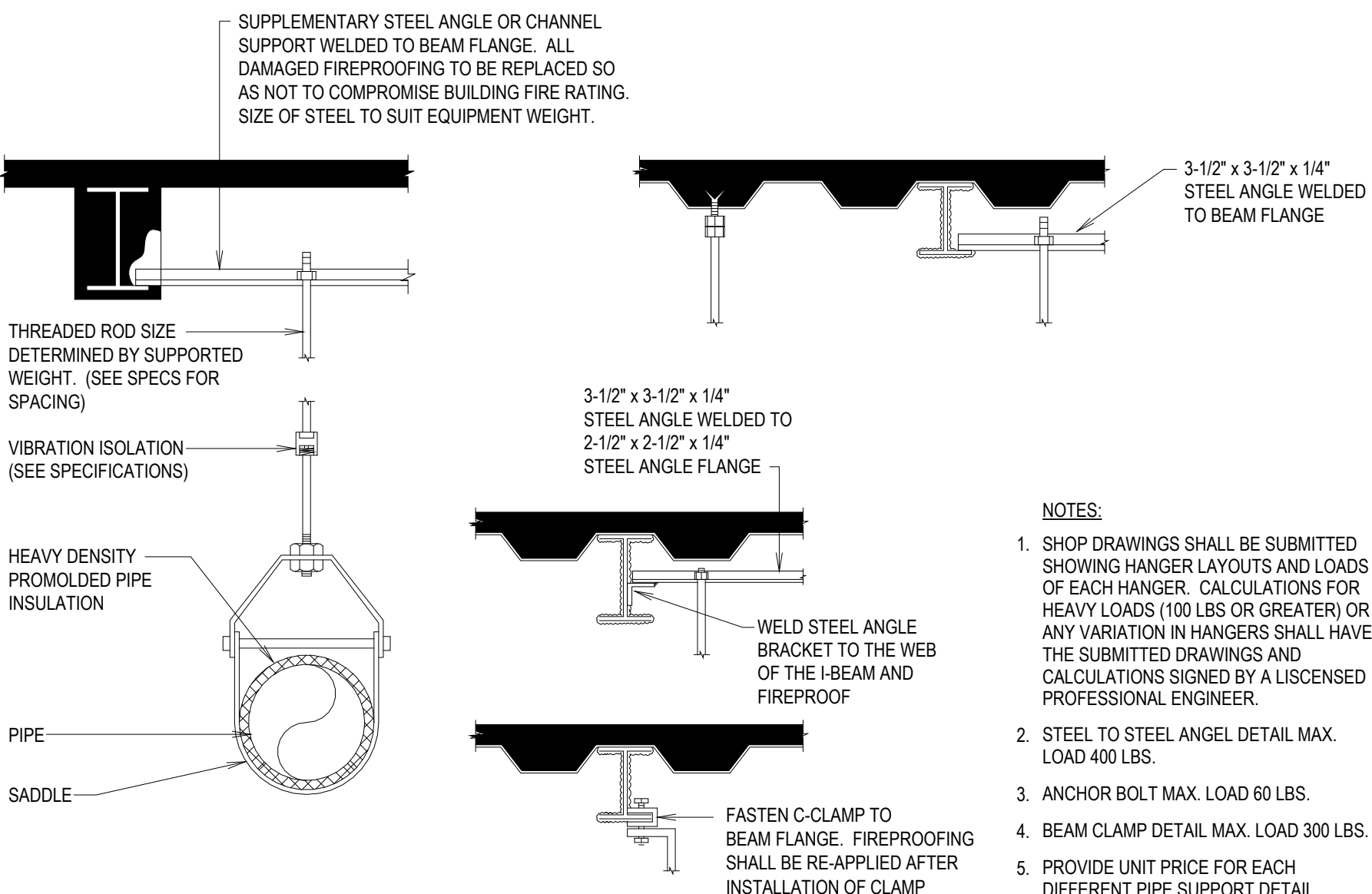
NOT TO SCALE



DETAILS OF LINEAR DIFFUSER INSTALLATION

NOT TO SCALE

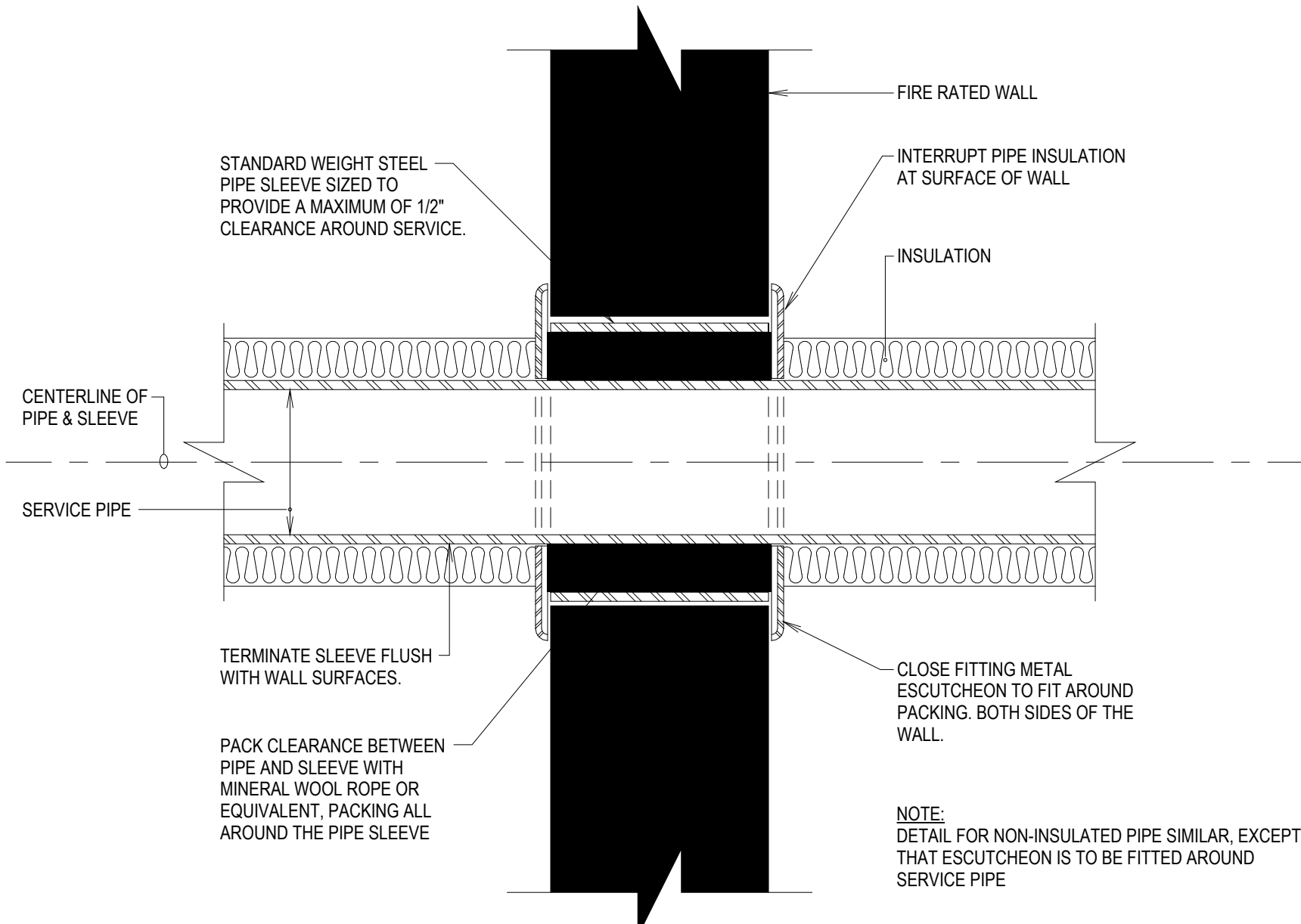
HD-047



TYPICAL PIPE HANGING DETAILS

NOT TO SCALE

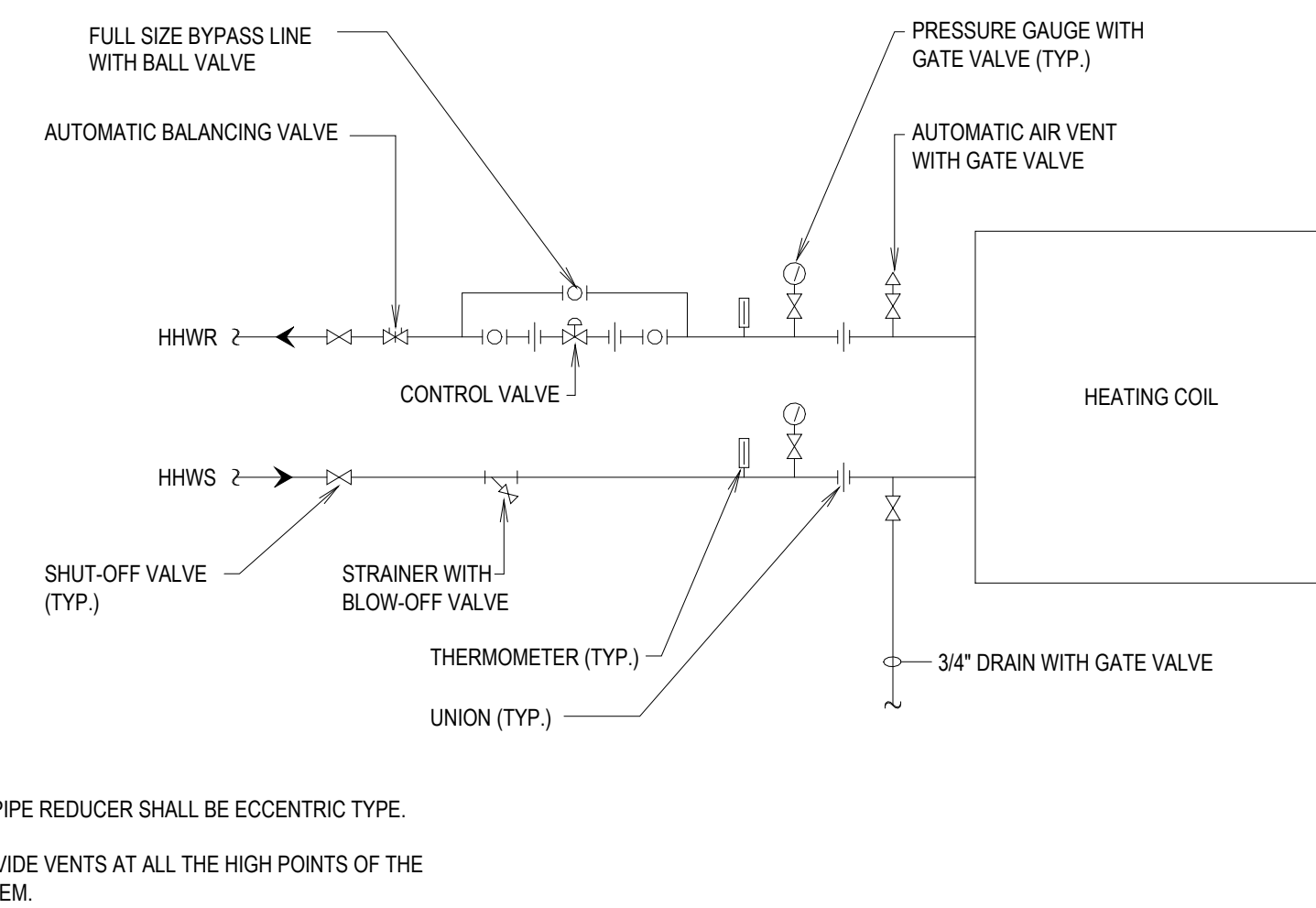
HD-036



PIPE THROUGH REQUIRED RATED PARTITION DETAIL

NOT TO SCALE

HD-063



HEATING COIL CONNECTION-HOT WATER (SINGLE COIL)

NOT TO SCALE

HD-078

MKDA

902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

ARCHITECT OF RECORD:

JOHN F. GIUSEFFI, ARCHITECT, PC.
902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

snowflake
7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:

AMA CONSULTING ENGINEERS
825 8TH AVE #18
NEW YORK, NY 10019
212.944.7722

AVIT/SEC CONSULTANT:

NORTHLAND CONTROLS
44150 S GRIMMER BLVD
FREMONT, CA 94538

FOOD SERVICE DESIGN CONSULTANT:

COLBURN GUYETTE
100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

DESIGN CONSULTANT:

JPC ARCHITECTS
909 112th Ave NE, Suite 206
Bellevue, WA 98004

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PROJECT:

SNOWFLAKE
7 TIMES SQUARE, NEW YORK, NY
FLOORS 29, 30, 31

DRAWING TITLE:

MECHANICAL DETAILS SHEET 2 OF 3

SEAL & SIGNATURE

DATE: 05.22.2026

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902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

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902 BROADWAY, NEW YORK, NY 10010 | 212.532.9800

CLIENT:

snowflake
7 TIMES SQUARE
NEW YORK, NY 10036

MEP CONSULTANT:

AMA CONSULTING ENGINEERS

825 8TH AVE #18
NEW YORK, NY 10019
212.944.7722

AVIT/SEC CONSULTANT:

NORTHLAND CONTROLS

44150 S GRIMMER BLVD
FREMONT, CA 94538

FOOD SERVICE DESIGN CONSULTANT:

COLBURN GUYETTE

100 LEDGEWOOD PL #104
ROCKLAND, MA 02370

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